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**SPECIFICATION FOR
SINGLE CIRCUIT OVERHEAD LINES
ON WOOD POLES FOR USE AT HIGH
VOLTAGE UP TO AND INCLUDING 33kV**

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Engineering Directorate

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SPECIFICATION FOR SINGLE CIRCUIT OVERHEAD LINES ON WOOD POLES FOR USE AT HIGH VOLTAGE UP TO AND INCLUDING 33 kV

FOREWORD

Light duty overhead lines to the Energy Networks Association Technical Specification (ENATS) 43-10 and heavy duty overhead lines to ENATS 43-20 have been used since the 1970s. The advent of the first issue of this Specification in 1988 brought about a change in the design basis for overhead lines and considered varying wind and ice loadings for a specific locality in a semi-probabilistic, load-factored manner.

The introduction of prEN 50XXX for lines up to and including 45 kV considers two design approaches, namely fully probabilistic and historic deterministic approaches, neither of which conforms to the approach used in Issue 1 of this Specification. This has created the need for this revised Specification to be issued.

1 SCOPE

prEN 50XXX covers the design requirements for overhead lines on wood poles for operation at high voltages up to and including 45 kV. Although included within this new standard, covered conductor lines are not considered in this Specification, and readers are directed to ENATS 43-121 in this respect.

Users of prEN 50XXX are offered a choice of design bases, and each National Committee within Europe is required to declare within the National Normative Aspects (NNAs) of the Standard the approach that will apply within their territory:

- (i) a deterministic design (called the “Empirical” design approach).
- (ii) a probabilistic design (called the “General” design approach).

The “General” design approach was not considered practical, nor economic, for use in the UK due to the requirement that intermediate supports should be designed to cater for the full effect of vertical loads, including heavy icing events in still air, and also in limiting the pole top deflection to less than 10 % of pole length under ultimate load conditions. Such extreme loading scenarios have not commonly been considered for wood pole line designs in the UK and as a result, the UK National Committee decided to adopt the “Empirical” design approach for new wood pole lines built to prEN 50XXX, and which would be similar to overhead line designs produced historically

It was recognized that the span/ spacing work previously included with the semi-probabilistic, load-factored design approach as detailed in the first issue of this Specification should be retained, as this had proven to give good conductor clash performance in practice. The original weather mapping has therefore been retained in this respect only.

In order that Users may select their preferred conductor size(s) and design basis and to avoid publishing the very large number of tables that would otherwise be required to accommodate all possible permutations in a written document similar to traditional specifications, this Specification is provided in part as a suite of computer spreadsheets (Microsoft Excel Workbook filename: 4340_EMPIRICAL.XLS).

Users are required to generate their own tables using the appropriate spreadsheet. These tables, together with the text from the written part of this Specification then form the User's own specification. A User Guide to the spreadsheet suite is provided as Appendix A to this Specification.

2 DEFINITIONS

Aeolian Vibration: Conductor oscillation caused by low velocity winds (1 m/s to 7 m/s) blowing steadily across the conductor.

Average Span: The arithmetic average of a number of spans in a line or section of line.

Auxiliary Equipment: Equipment other than that forming part of the line design that may be erected on supports such as transformers, fuse-gear etc.

Basic Span: The span length adopted for sag / tension calculations.

Clashing: The electrical contact between two dissimilar line conductors of an overhead line displaced from their normal position by environmental forces such that the conductors are likely to touch.

Conductor Creep: Permanent long term elongation of the conductor.

Conductor Downpull: The vertical loading imposed by the conductors corresponding to a gradient measured between adjacent pole tops.

Crossarm hamper: A descriptive term used to include all pole top crossarm components and fixings.

Everyday Tension: The design stress in an unloaded conductor at 5°C. Used in sag / tension calculations to limit harmful conductor vibrations.

Freezing Point Tension: The design tension of an unloaded conductor at 0°C in still air.

Ice Co-ordinate: The intercept on the ice axis of the weather incidence load line (see ENATR 111) whose value is given for differing height and UK location in the associated weather maps.

Intermediate Support: A support in a straight run of line on which the conductors are supported on pin insulators.

Maximum Conductor Pressure (MCP): The maximum transverse component of applied conductor load when subjected to a wind pressure.

Maximum Conductor Weight (MCW): The maximum vertical component of applied conductor load, including the weight of accreted ice, if present.

Maximum Span: The maximum permitted length of any span.

Maximum Working Tension (MWT): The absolute maximum conductor tension assessed at -5.6°C with wind and ice loading.

Minimum Failing Load (MFL): The minimum applied mechanical load that will cause failure by breakage of a component.

Overtensioning: Excess tension applied above normal tension at time of erection to compensate for conductor creep.

Pin Angle Support: A support at which a line deviates and the conductors are supported on pin insulators.

Pre-tensioning: The tension treatment applied to a conductor for a short duration before final erection tension is established to remove a proportion of conductor creep.

Rated Breaking Strength (RBS): The rating assigned to a component (usually conductors) that defines the maximum mechanical load that the component will withstand without breakage.

Recommended Span: The average span length in any section to which the line should be planned. Individual spans will normally be within $\pm 20\%$ of the chosen basic span.

Sag: The vertical distance, under any system of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured at mid-span.

Section Angle Support: A support at which a line deviates and the conductors are made off on either side of the crossarm on tension insulator sets.

Section Support: A support in a straight run of line on which the conductors are made off on either side of the crossarm on tension insulator sets.

Span: The horizontal distance between adjacent supports. Individual spans will normally be within 20 % of the chosen basic span.

Wind Co-ordinate: The intercept on the wind pressure axis of the weather incidence load line (see ENATR 111) whose value is given for differing heights and UK locations in the associated weather maps.

Weather Zone: A geographical area in which the likely mean wind pressure and absolute maximum ice accretion thickness may be described by a numeral and letter respectively. The wind co-ordinate is described in 190 N/m² increments, whilst the ice co-ordinate is measured in 10 mm diametric thickness increments.

Wind Loading Span: Half the sum of the spans adjacent to the support.

3 REFERENCES

This Specification makes reference to the following documents:

The Electricity (Overhead Lines) Regulations, 1970, SI 1355. **(NOW WITHDRAWN)**

The Electricity Safety, Quality and Continuity Regulations 2002, SI 2665.

BS 1990 Part 1	Wood poles for overhead lines (Power and telecommunications lines).
BS 3288	Insulator and conductor fittings for overhead power lines.
BS 5950 Part 1	Structural use of Steelwork in building – Code of Practice
BS 7884	Copper and copper cadmium stranded conductors for overhead electric traction and power transmission systems
BSEN 20898-1	Mechanical properties of fasteners - Bolts, screws and studs
EN 12465	Wood poles for overhead lines – Durability
EN 12479	Wood poles for overhead lines – Methods of measurement and permissible deviations
EN 12509	Wood poles for overhead lines – Test methods
EN 12510	Wood poles for overhead lines – Strength grading criteria
EN 12511	Wood poles for overhead lines – Determination of characteristic values
prEN 14229	Wood poles for overhead lines – Requirements
EN 50182	Conductors for overhead lines – round wire concentric lay stranded conductors
EN 50341 Part 1	Overhead electrical lines exceeding AC 45 kV – General requirements Common specifications
prEN 50XXX-1	Overhead Electric Lines with Rated Voltages Exceeding AC 1 kV up to and including AC 45 kV – General Requirements Common specifications
ENATS 43-10	11 kV single circuit overhead lines of light duty construction on wood poles.
ENATS 43-20	11 kV and 33 kV single circuit overhead lines of heavy-duty construction on wood poles.
ENATS 43-88	Selection and treatment of wood poles and associated timber for overhead lines
ENATS 43-90	Anti-climbing devices for HV lines up to and including 400 kV.
ENATS 43-91	Stay strands and stay fittings for overhead lines.
ENATS 43-93	Line insulators.
ENATS 43-95	Steelwork for overhead lines.
ENATS 43-96	Fasteners and washers for wood pole overhead lines.
ENAEER L38/1	Protective Greasing of PVC-covered copper and bare and PVC covered aluminium based overhead line conductors.

4 DESIGN DATA

The loading requirements of the Electricity (Overhead Lines) Regulations 1970 (now withdrawn) have been reviewed and updated / amended to reflect more onerous weather conditions at higher altitudes. The revised loading cases are now:

“*Normal*” altitudes: Wind pressure 380 N/m^2 with a radial ice thickness of 9.5 mm. For conductors up to and including 35 mm^2 copper equivalent area (60 mm^2 aluminium based conductors), an alternative loading is possible with wind pressure of 760 N/m^2 on bare (i.e. non-iced) conductors. This latter loading has proven to give satisfactory performance over many years and within many areas of Great Britain and Northern Ireland. Since the acceptability of the loading is dependent on site topography, location, meteorological data and security of supply, the requirement for its use in a specific environment will be determined by the local distribution network operator.

“*High*” altitudes: Wind pressure 570 N/m^2 with radial ice thickness 12.5 mm. For conductors up to and including 35 mm^2 copper equivalent area (60 mm^2 aluminium based conductors), an alternative loading is possible with wind pressure of 760 N/m^2 on bare (i.e. non-iced) conductors as detailed above.

In the contexts, “*Normal*” altitude is defined as all of Great Britain, except Scotland, site altitudes not exceeding 300 m. For Scotland, site altitudes not exceeding 200 m. More onerous requirements, perhaps due to known localised wind situations may be additionally specified. “*High*” altitude is defined as all locations up to 500 m. For lines at altitudes greater than 500 m, a special consideration using local meteorological data should be made.

The following ice / snow densities are considered depending on locality and altitude: *glaze ice* 913 kg/m^3 , *wet snow* 850 kg/m^3 and *rime ice* 510 kg/m^3 . The User must be familiar with the worst icing mechanism likely in a locality in order to accurately calculate the expected structure and conductor loadings. Where such data is unavailable, it is recommended that the User adopts *glaze ice* density within the SAGNTEN spreadsheet.

4.1 Conductors

Where the non-iced loading is employed for conductors of cross sectional area up to and including 35 mm^2 copper equivalent area (60 mm^2 aluminium based conductors), the minimum factor of safety shall be 2.5 on their nominal breaking load. The minimum factors of safety on all conductors subject to an ice load shall be 2.0 on their nominal breaking loads. This limitation in conductor tension is known as Maximum Working Tension (MWT) and occurs at minimum conductor temperature, taken as -5.6°C as used in the SAGNTEN spreadsheet.

Conductors have self-damping characteristics that reduce the amplitude of damaging aeolian vibrations. The self-damping effect increases with conductor size, but decreases as strands become locked more tightly together with increasing tension or compaction. Experience has shown that the self-damping of normally bare aluminium based conductors requires a maximum Everyday Tension (EDT) of 20 % of Rated Breaking Strength (RBS) at 5°C to reduce aeolian vibration to an acceptable level in open terrain. For copper based conductors, the EDT limit is normally 33 % of RBS, also at 5°C .

4.2 Conductor Spacing to avoid Clash

Maximum span lengths to avoid clash for any conductor and associated tension regime may be calculated using the COND SPACING spreadsheet. The method employed is as the previous issue of this specification using assumed smoothed gust and lull wind pressures of 1.832 times the mean- and 0.546 times the mean- wind pressures respectively for a gust factor of 1.5. Further information regarding the philosophy is contained within ENATR 111. The likely mean wind and ice accretion levels for a specific locality and elevation are indicated as “weather zones” on the maps included in section 8 of this specification. The numeral indicates the wind pressure in 190 N/m² increments and the letter as 10 mm diametric ice thickness increments for the indices A through to E respectively. The worst combination of wind and ice loading is used for the

calculation, with the actual wind and ice loads employed enhanced by a withstand factor of 1.1.

It is recommended for lines at “Normal” altitudes that the conductor spacing be related to a “2B” weather zone, whilst for “High” altitudes the conductor spacing be related to a “3C” weather zone. The User may employ other weather zones as indicated by the mapping, although it is recommended that less onerous weather zones than those noted above for “Normal” and “High” altitudes should be avoided.

4.3 Crossarm Assemblies

Typical crossarm assemblies have been designed in accordance with BS 5950 Part 1 using a design stress equivalent to the ultimate stress divided by a factor of safety of 2.5 that shall not be exceeded for all likely loading conditions (i.e. for mild steel, allowable stress $430 \div 2.5 = 172 \text{ N/mm}^2$). The steelwork sizes indicated on the structure drawings follow those historically included in the previous issue of this Specification and detailed in ENATS 43-95.

As with the previous issue of this Specification, it is assumed that for single pole structures equipped with crossarm struts, the vertical loads will be reacted by the pole bolts as follows: one third of total vertical load by the top pole bolt and two thirds of total vertical load by the bottom pole bolt.

Users must confirm the suitability of a particular crossarm / conductor/ downpull / maximum span length combination they intend to adopt by employing the Excel Workbook and checking the SINGLE POLE FIXING or H POLE FIXING spreadsheets as appropriate in conjunction with the CROSSARM MEMBER and CROSSARM STRUT spreadsheets.

4.4 Supports

The stresses created in intermediate supports are considered as bending stresses only due to wind load on conductors and insulators, the loading point being approximately 160 mm above pole tops for 11 kV lines. (Note: this dimension is the standard for 11 kV construction and equates to 260 mm above the top flange of the crossarm, but may need to be increased for differing insulator dimensions and / or higher voltages).

The stresses created in stayed supports are crippling stresses due to stay tension, conductor weight / downpull and crossarm hamper loads acting at the pole top. The

distribution of stresses in the limbs of “H” supports on which stays are equally deployed on the two poles and are adjusted to equal tension is considered to be 50 % of stress in each limb. Unequal staying (two stays on one limb and one on the other) is included in the spreadsheet H STRUT, if required by the User. In this instance the strut load is assumed to be 60 % or 40 % respectively.

The strength of supports is derived from formulae contained in BS 1990 part 1 and in respect of *Pinus Sylvestris* wooden products; the pole producers in the UK have confirmed the following parameters:

Mean ultimate extreme fibre stress	53.3 N/mm ²
Average modulus of elasticity	10 054 N/mm ²

These parameters are based on a wood pole population whose southern border occurs at 60° latitude. Where some other species and / or population is considered, other strength parameters may be applicable and in this respect, amended values should be employed in the associated spreadsheets, based on corroborating data from the pole producers.

4.5 Fasteners

Dimensionally in accordance with ENATS 43-96 and with allowable stresses based on the ultimate tensile strength detailed in BS EN 20898-1 in respect of mild steel bolts (Grade 4.6) (i.e. $400 \div 2.5 = 160$ N/mm² maximum axial stress on net bolt cross sectional areas for all arrangements and $400 \div 2.5 \times 0.75 = 120$ N/mm² maximum shear stress on (i) net bolt cross sectional areas for single pole terminal crossarm and all double crossarm arrangements and (ii) gross bolt cross sectional areas for all other arrangements). High Tensile Steel bolts (Grade 8.8) have an increased allowable axial stress of 320 N/mm² and increased allowable shear stress of 240 N/mm², although this grade of bolt is rarely used for wood pole lines in the UK.

5 MATERIALS

Unless otherwise specified, all components shall comply with the appropriate British Standard, BS EN or ENA Technical Specification or equivalent.

5.1 Anti Climbing Devices & Safety Signs

In accordance with ENATS 43-90.

5.2 Conductor

The conductor shall be in accordance with BS EN 50182 in respect of aluminium-based conductors, and BS 7884 in respect of copper conductors. Conductors are to be greased in accordance with the requirements of ENAER L38/1 before use in the line. The weight of the conductor should be augmented to include for the mass of grease for sag / tension calculations. *Note: aluminium-based conductor parameters to BS EN 50182 and copper conductor parameters to BS 7884 (as used in the Excel Workbook) vary slightly from those parameters previously included in the now superseded standards BS 3242, BS 215 and BS 125 standards respectively.*

5.3 Conductor Terminations, Joints and other Fittings

In accordance with ENATS 43-92.

5.4 Fasteners and Washers

In accordance with ENATS 43-96 and BS EN 20898-1.

5.5 Insulators and Associated Fittings

In accordance with ENATS 43-93 and BS 3288. The use of polymeric insulators is not precluded by this Specification.

5.6 Stays and Wood Baulks

In accordance with ENATS 43-91.

5.7 Steelwork

Generally in accordance with ENATS 43-95, and designed to BS 5950: Part 1.

5.8 Wood Poles

Wood poles shall be fabricated in accordance with prEN 14229, BS EN 12465, BS EN 12479, BS EN 12509, BS EN 12510, BS EN 12511, and ENATS 43-88 Parts 1 and 2. For "H" poles, the minimum pole top diameters for Medium-grade poles is 175 mm and for Stout-grade poles is 210 mm, as indicated on drawing 434002.

6 TECHNICAL REQUIREMENTS

6.1 Span Lengths

The Basic/ Recommended span used for a particular conductor may be varied as required, based on the average span length of the line. Typically, a Basic Span will be in the range of 80 m to 130 m. Use of different basic spans within one overhead line is only approved where effective back staying, or other means, are provided to stabilise the out of balance tensions between the different sections at the transition structure.

In keeping with the earlier Issue 1 of this Specification, maximum span length suitability can be checked using the COND SPACING spreadsheet which includes the original ENATR 111 / ENATS 43-40 withstand data based on the calculated Freezing Point Tension (FPT) for the specified conductor. The maximum span can be calculated from the spreadsheet using a User specific equivalent spacing that must be less than the actual phase spacing at the crossarm. In most cases, the maximum span is within +20 % of the basic span length.

Where the combination of conductor / weather zone / basic span indicates span lengths that are considered too short, it is recommended that either a stronger conductor or shorter Basic span be employed.

6.2 Failure Containment Policy

Where a line is to be built in a particularly exposed location and/ or an enhanced security of supply is required, the following arrangements are considered appropriate:

- (1) Intermediate supports to 434003 fitted with failure containment devices.
- (2) Intermediate supports to 434004.
- (3) Pin angle arrangements to 434004 at least equipped with twin splayed stays spread at 45° to the pole. *Not recommended for higher security feeders.*
- (4) Section supports to 434005 of a pole grade at least able to support 125 % of the windspan that applies in practice unless the pole grade is Stout, in which case this grade shall be adequate to satisfy this clause. Stays to be employed on both sides of the structure in the plane of the line.
- (5) Any of the "H" section, section angle or terminal structures installed in accordance with this specification. Stays to be employed on both sides of the structure in the plane of the line for section structures.

6.3 Wind Loading Spans

Tables detailing wind-loading spans for intermediate and section supports are User-generated by the Excel Workbook in the WOODPOLE spreadsheet. Pole sizes are shown defaulted to the minimum sizes quoted in BS 1990: Part 1, although Users may elect to employ their own specific pole sizes.

The BS 1990: Part 1 range of minimum / maximum pole sizes is given within this spreadsheet for guidance.

In respect of intermediate structures only, where vertical loads have always been ignored, the factor of safety on the pole when subjected to transverse loads will be increased with increasing declination (ie. Factor of safety = 2.5 for level conditions up to 1 : 10 declination; factor of safety = 3.0 for 1 : 7.5 declination and factor of safety = 3.5 for 1 : 5 declination). The appropriate factor of safety will be inserted in the DESIGN PARAMETERS spreadsheet for unstayed poles. Such an increased loading criteria will mitigate, to some extent, the effect of vertical loading on these structures.

6.4 Maximum Angles of Line Deviation

The maximum angles of line deviation on pin insulators or with tension insulator sets can be determined from the SINGLE POLE FIXING or H POLE FIXING spreadsheets as appropriate, based on the crossarm and strut data entered in the CROSSARM MEMBER and CROSSARM STRUT spreadsheets.

The recommended maximum angle of line deviation on a structure with tension insulator sets is 60°.

6.5 Stay Arrangements

Tables detailing staying arrangements are User-generated by the supplied spreadsheets S STAYS and H STAYS as appropriate.

'H' poles erected as terminals, or out of balance stayed sections between sections of lines built to dissimilar basic spans, may be equipped with single, double or in some cases triple stays per leg. Double stays are always to be splayed with a minimum spacing as indicated in the STAY FOUND spreadsheet depending on the stay angle employed.

Triple stays shall comprise two splayed stays as above set at the approved angle to the pole, the third stay being placed to bisect the two and entering the ground at least one metre behind them viewed from the pole.

In extreme weather conditions, stayed supports may be required to afford appreciable failure containment capability. In practice the design loading when applied to the case of large angles of line deviation and terminals, is generally more onerous than abnormal cases associated with broken conductors when applied to the same structures with correctly installed and maintained stays. A correctly stayed 'H' structure is deemed to provide an acceptable level of failure containment when used up to the ultimate loading case.

In the case of small angles of line deviation, single stays set at the minimum angle required to resist the normal design loading case do not offer significant failure containment capability.

When out-of-balance tension exists on a section pole, balancing stays should be fitted as required, the design for which will need to be assessed by Users separately.

6.6 Conductor Design/Erection Sags and Tensions

Tables detailing the ordinates for plotting uplift and ground clearance curves are User-generated by the supplied spreadsheet SAGNTEN and are based on conductor design tensions.

Tables detailing conductor design and erection sags and tensions are User-generated by the supplied spreadsheet SAGNTEN.

6.7 **Strut Loading of Supports**

Supports shall be in accordance with the general arrangement drawings that form part of this Specification. It is recommended that all stays on angle and terminal poles be set at the maximum stay angle possible to afford the structure the greatest level of failure containment capability.

Tables detailing the strut loading imposed on angle and terminal supports are User-generated by the supplied spreadsheets S STRUT and H STRUT as appropriate. The loading is then compared with the strut loading capability of the pole size required using the supplied spreadsheet WOODPOLE. The strut loading calculated **must** be less than the pole strength capability indicated by the WOODPOLE spreadsheet and the bearing capacity of the soil as indicated by the STRUT FOUND spreadsheet.

Where a single stay is employed on an "H" pole structure (ie. "H" pole occasional long span 434007 with a very small angle of line deviation), the strut capacity of one limb of the structure must be greater than the vertical component of stay tension due to the stay plus half the vertical loading due to conductors, steelwork, etc.

6.8 **Pole Foundations**

The strut loading obtained from the supplied spreadsheets S STRUT or H STRUT as appropriate must be less than the respective value for soil bearing strength obtained from the supplied spreadsheet STRUT FOUND. An average soil bearing capacity of 429 kN/m² (4 tons/ft²) has been used within this spreadsheet.

The supplied spreadsheet WOODPOLE details the requirement for foundation baulks on intermediate and section supports and considers three soil grades for lateral capacity: 628 kN/m², 471 kN/m² and 314 kN/m² per metre depth for good, good / average and average / poor soils respectively. These capacities are as assumed in Issue 1 of this specification.

6.9 **Minimum Height of Line Conductors and Clearance**

The minimum height of line conductors shall comply with the Electricity Safety, Quality and Continuity Regulations 2002.

6.10 **Provision of safety signs**

Each support shall be provided with at least one safety sign to ENATS 43-90. The sign shall be mounted approximately 3 m above the ground level, above any anti-climbing guard and clearly visible to an observer on the ground.

6.11 Precautions against access

The types of support requiring precautions against access and the methods to be employed are detailed in ENATS 43-90.

6.12 Erection of Auxiliary equipment

No allowance has been made in the basic support consideration for the additional mechanical loadings that would be imposed by auxiliary equipment and additional conductors / cables. Where such an additional is required (i.e. the additional weight due to pole mounted transformer), this may be added in the LINE PARAMETERS spreadsheet.

6.13 Relationship with other bodies

Where lines to this standard are erected over or alongside the plant of Telecommunications Operators or Network Rail, then the provisions of the relevant joint agreements shall be complied with. Additionally, the requirements of Waterways Authorities should be noted.

7 CONDUCTOR ERECTION GUIDANCE

7.1 Conductor Creep

Suspended conductors are subject to longitudinal stresses that cause permanent long-term elongation termed conductor creep. This results in an increase in sag that must be accounted for by the User of this Specification, in order to ensure satisfactory ground clearance exists throughout the life of the line.

Elongation due to creep has two components:

- (i) Metallurgical creep - related to the time for which a load is applied.
- (ii) Geometrical settlement - related to the magnitude of the load and the stresses at strand crossover points as the strands of the conductor tighten together and deform.

For a given conductor the total amount of elongation depends on the erection tension and the mechanical and thermal conditions that occur in service. By forecasting the anticipated conductor service conditions, it is possible to predict the resulting elongation either by the use of a suitable creep predictor formula, or by extrapolating the results of conductor creep tests.

Compensation for the effects of conductor creep can be achieved by using one or more of the following three techniques:

- (i) Pre-tensioning
- (ii) Over tensioning
- (iii) Ordinate shift

7.2 Pre-Tensioning

Prior to final sagging, a conductor can be pre-tensioned to reduce the amount of elongation that will occur in service. Pre-tensioning conductors has the additional advantage of proof testing other line components. For safety reasons, the pre-tension value should not exceed any of the following: MWT or 50 % of the conductor rated breaking strength (RBS).

A typical pre-tensioning regime would consist of the conductor being tensioned to this value for one hour. At 15 minute intervals during this period, the tension should be adjusted to maintain the pre-tension value. At the end of the period, the conductors should be tensioned in accordance with the appropriate erection tension table and then terminated.

7.3 Over-Tensioning

A conductor can be erected at an increased tension and then terminated. The reduction of sag due to the increased tension compensates for the increase in sag due to conductor creep during the life of the line. The amount of over tension applied can be expressed as a percentage increase over the design tension or alternatively as an equivalent temperature shift. The SAGNTEN spreadsheet enables either method to be applied to the calculated design tension tables in order to produce the erection tension tables.

Typical overtension values used historically are 5 % or 10 % at 15°C depending on the conductor type and construction. Recommended temperature shift values for a specific conductor(s) should be sought from the conductor manufacturers.

When using either over-tensioning method, it is necessary to consider the possible short-term vibration and design limitations of other components, especially early in the life of the overhead line.

7.4 Ordinate Shift

A conductor can be erected with additional ground clearance to compensate for the additional sag caused by conductor creep, although such a method will result in taller structures and the maximum span to avoid clash will be reduced due to the increased sag.

The SAGNTEN spreadsheet allows the User to modify template ground clearance ordinates by reducing the tension either by temperature shift or percentage reduction. Such percentage reduction values at 15°C used historically for a 25-year creep period include:

Copper conductors	5 % reduction
Aluminium alloy conductors 50 mm ² csa	5 % reduction
Aluminium alloy conductors 100 and 175 mm ² csa	10 % reduction
ACSR conductors 175 mm ² (18 + 1) csa	20 % reduction
ACSR conductors 175 mm ² (37 + 1) csa	5 % reduction

7.5 General

A combination of creep compensation methods may be necessary depending on the stranding of the conductor. Recommendations may be obtained from conductor suppliers for specific conductor sizes, basic span and tension regimes.

8 WEATHER CO-ORDINATE MAPS

The following maps show ice and mean wind pressure co-ordinates applicable throughout the UK at site levels of between 0 m and 500 m. Data is not available for higher site locations, and clash performance calculations must be based on the data for 500 m level and adjusted in the light of local experience.

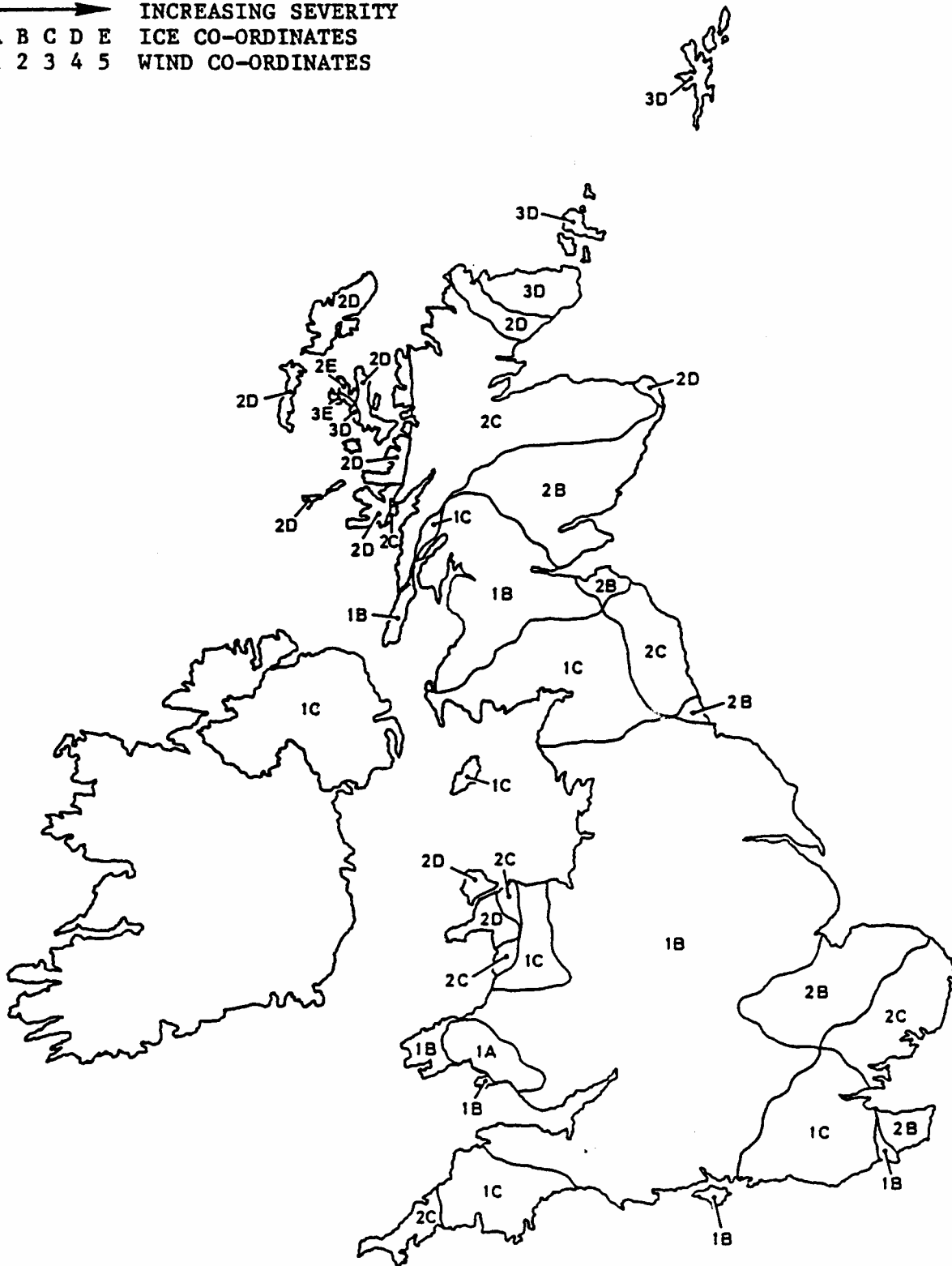
The five maps show the combined mean wind pressure and ice co-ordinates for “Weather Zones” at site location heights of:

0 m to 100 m
100 m to 200 m
200 m to 300 m
300 m to 400 m
400 m to 500 m

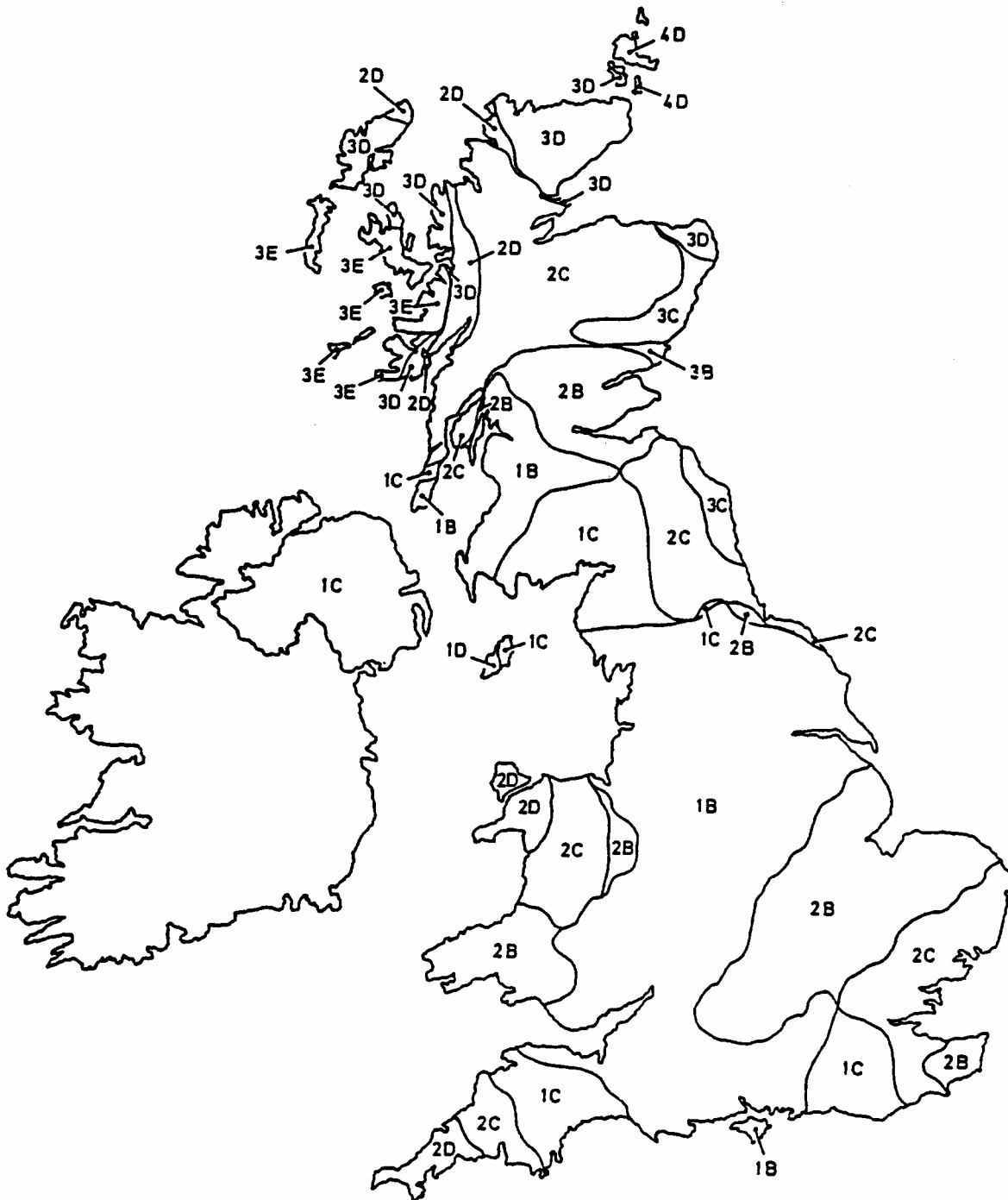
The weather mapping has been reproduced from Issue 1 of this Specification, but is limited within this Issue to conductor clash assessments only.

In this respect, it is recommended that a weather zone of “2B” be used for lines located at “*Normal*” altitudes, and weather zone “3C” be used for lines located at “*High*” altitude. Where a line is to be erected in a more severe weather zone, the applicable weather zone from the mapping should be used for conductor clash assessments.

—————→ INCREASING SEVERITY
 A B C D E ICE CO-ORDINATES
 1 2 3 4 5 WIND CO-ORDINATES

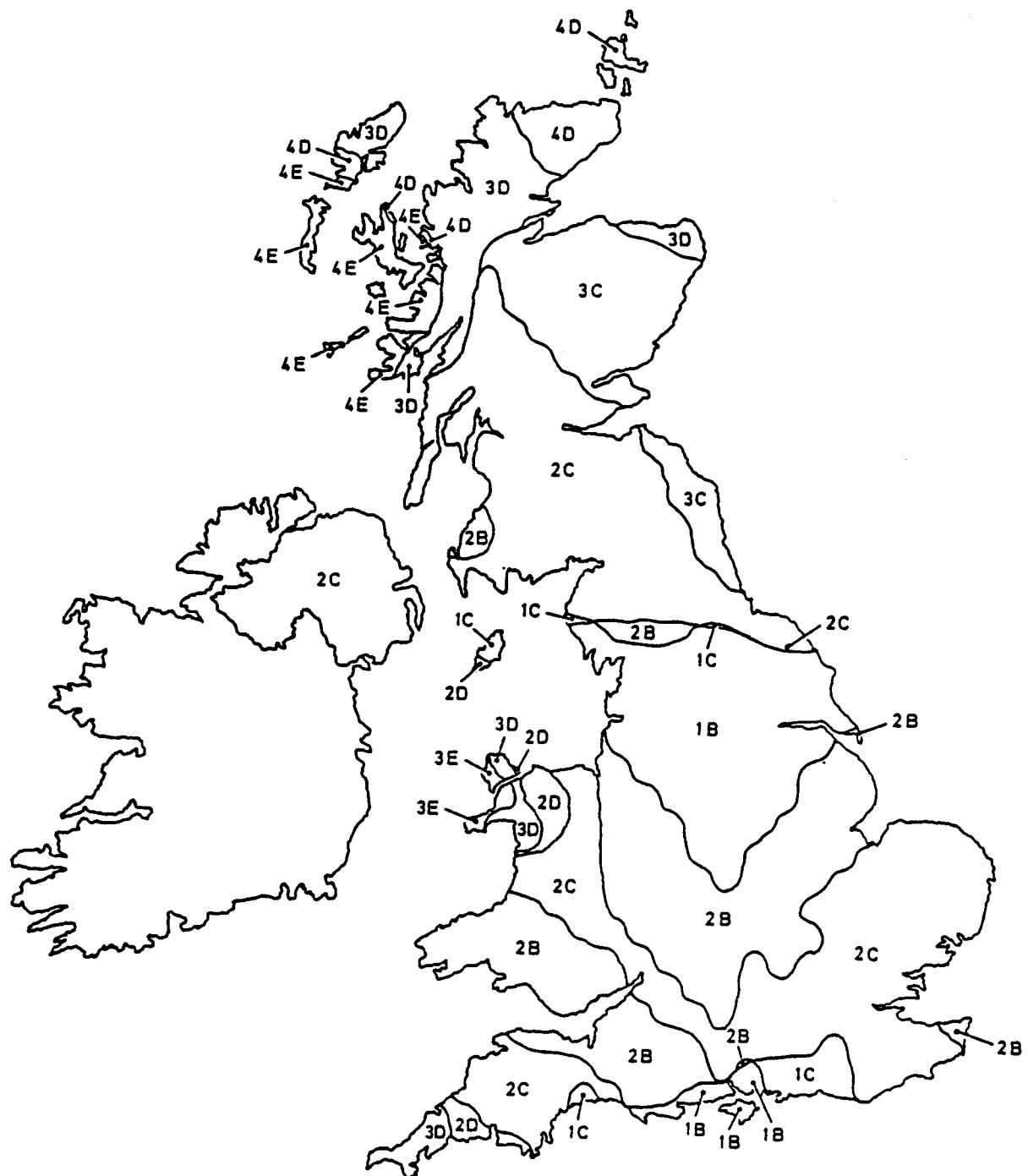


Weather Zones for site heights of 0 m to 100 m



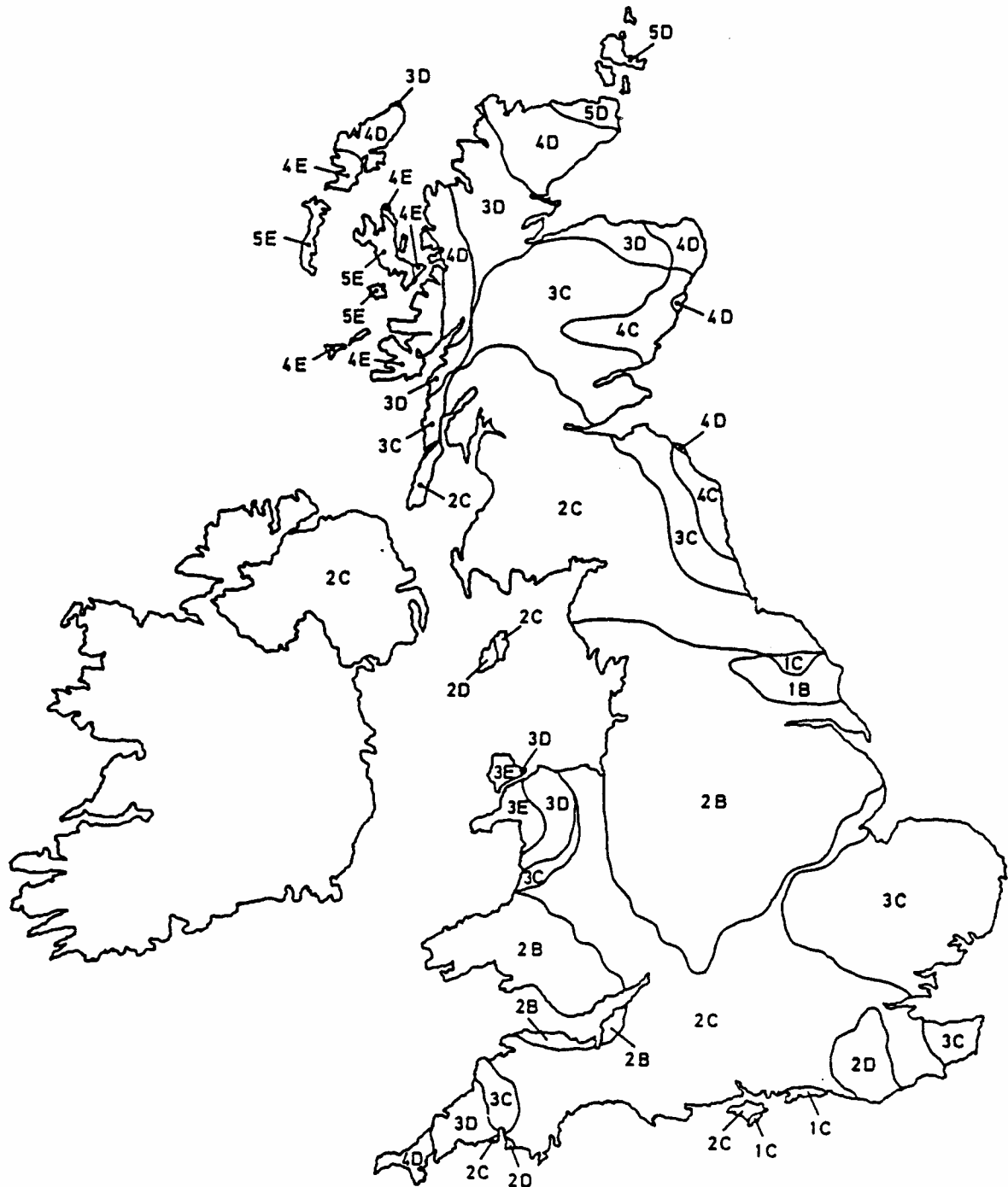
Weather Zones for site heights of 100 m to 200 m

→ INCREASING SEVERITY
 A B C D E ICE CO-ORDINATES
 1 2 3 4 5 WIND CO-ORDINATES



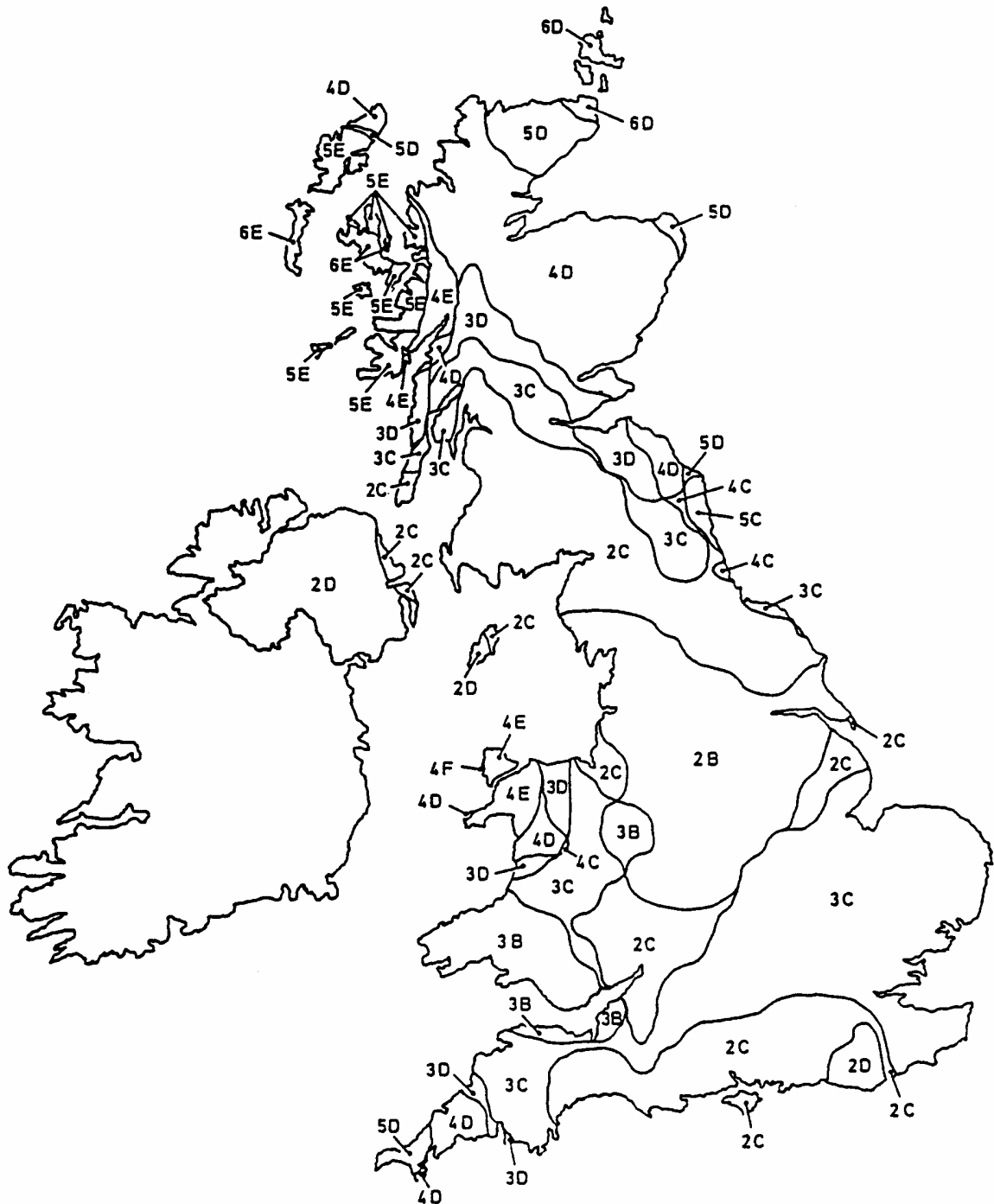
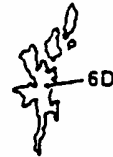
Weather Zones for site heights of 200 m to 300 m

—————→ INCREASING SEVERITY
 A B C D E ICE CO-ORDINATES
 1 2 3 4 5 WIND CO-ORDINATES



Weather Zones for site heights of 300 m to 400 m

—————→ INCREASING SEVERITY
 A B C D E ICE CO-ORDINATES
 1 2 3 4 5 WIND CO-ORDINATES



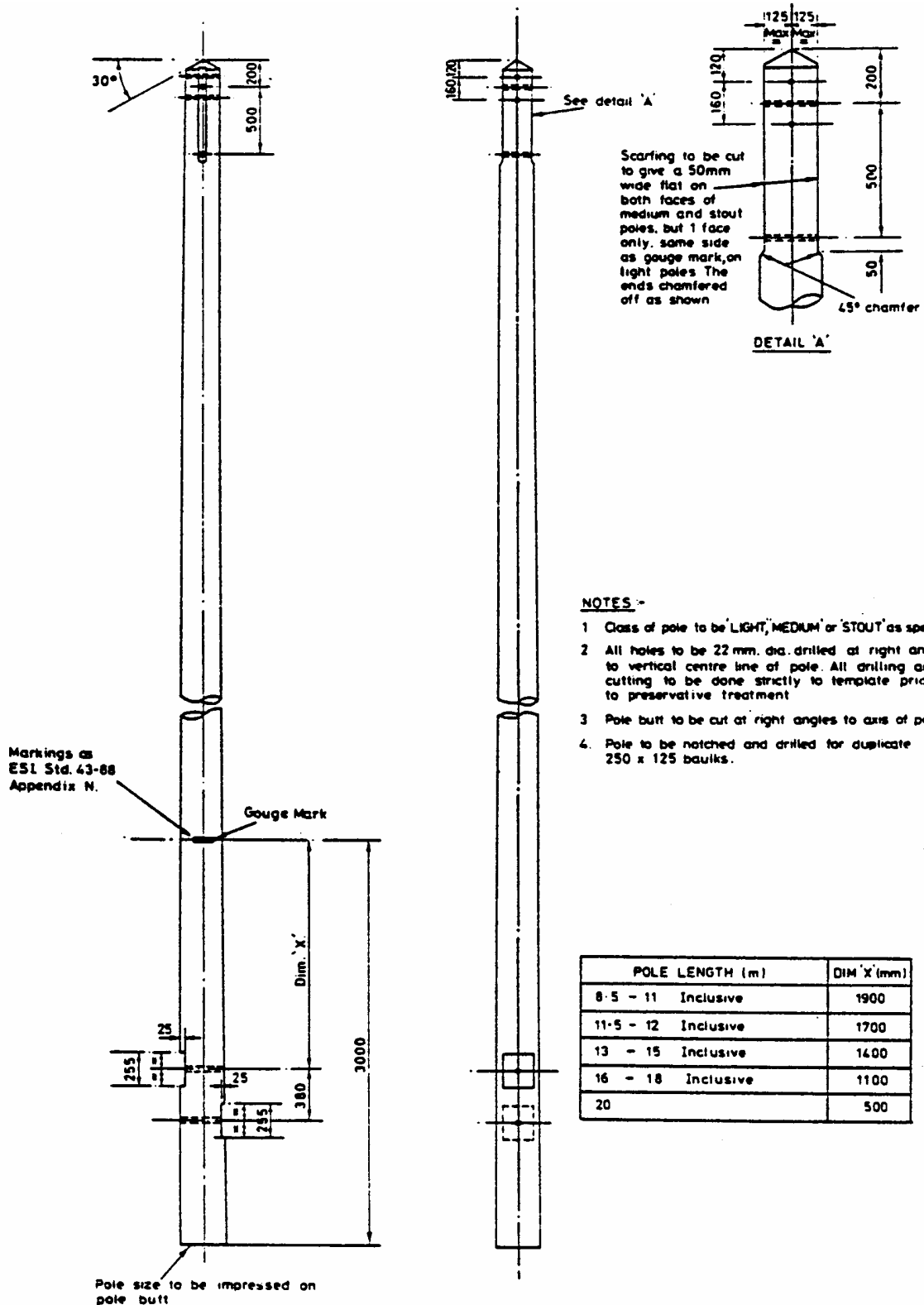
Weather Zones for site heights of 400 m to 500 m

9 GENERAL ASSEMBLY DRAWINGS and MATERIAL LISTS

This section gives material lists and general arrangement drawings for the different overhead line structures covered by this specification. A brief description of each support application is detailed in the table below, cross-referenced to the support drawing number.

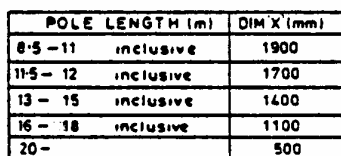
The applications for each support are given as a general guide only, the suitability of each support for use at a particular location being dependant on successful calculations using the Excel Workbook.

Drawing Number and Brief Description	Application
434001 Single pole fabrication	All Single pole structures
434002 "H" pole fabrication	All "H" pole structures
434003 Single Crossarm, Single Pole	Straight Line, small deviation angle, 1 : 10 downpull
434004 Double Crossarm, Single Pole	Straight Line, small deviation angle, 1 : 10 downpull
434005 Double Crossarm, Single Pole	Straight Line and Section, medium deviation angles, 1 : 10 downpull
434006 Double Crossarm, "H" Pole	Section, heavy deviation angles, 1 : 10 downpull
434007 Double Crossarm, "H" Pole	Section, occasional long span, heavy deviation angles, 1 : 10 downpull
434008 Tee-off, double crossarm, single pole	Tee-off
434009 Unbraced double crossarm, single pole	Termination
434010 Double crossarm, single pole	Termination
434011 Double crossarm, "H" pole	Termination
434012 Single crossarm, single pole	Renovation of existing poles using existing conductors, straight line, limited deviation angles



All dimensions are in millimetres unless otherwise stated

Drawing 434001 Single Pole Fabrication



NOTES -

1. Class of pole to be 'MEDIUM' or 'STOUT' as specified
2. 'MEDIUM' class poles to have minimum dia. of 175mm at top.
'STOUT' class poles to have minimum dia. of 210mm at top.
3. Poles to be matched
4. All holes to be 22mm dia drilled at right angles to vertical centre line of pole. All drilling and cutting to be done strictly to template prior to preservative treatment
5. Pole butt to be cut at right angles to axis of pole
6. Pole to be notched and drilled for duplicate
250 x 125 baulks

All dimensions are in millimetres unless otherwise stated

Drawing 434002 "H" Pole Fabrication

**Materials List for Drawing No. 434003
Intermediate or Pin Angle Pole (Single Crossarm Member)**

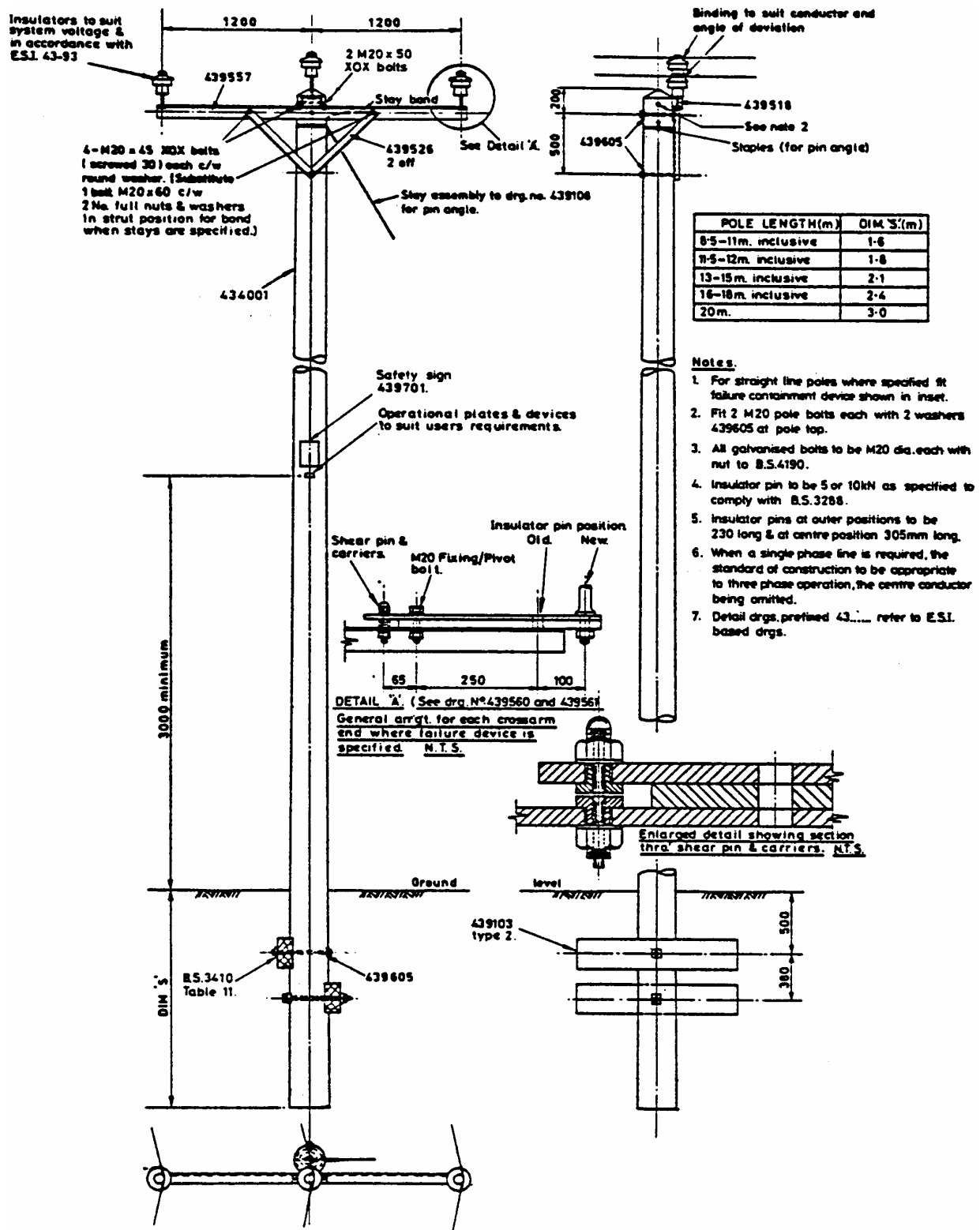
Quantity	Description	ENATS Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
1	Wood Pole	4312101	4 (pole)	
2	Kicking Block	439103 Type 2	2 (pole)	
1	Crossarm Member	439557	-	
2	Crossarm Strut	439526	2	4
1	Insulator Bracket	439518	2	
2	Square Flat Washer	439604	-	
6	Square Curved Washer	439605	-	
3	11 kV Pin Insulator 20 kV Pin Insulator 33 kV Pin Insulator 33 kV Line Post Insulator	439301 439302 439303	- - - -	
2	11 kV Pin 230mm 5 kN	BS 3288 Ref. 28		+
	11 kV Pin 230mm 10 kN	BS 3288 Ref. 29	-	
1	33 kV Pin 305mm 5 kN	BS 3288 Ref. 30		
	33 kV Pin 305mm 10 kN	BS 3288 Ref. 31	-	
1	Safety Sign	439004	-	

Notes:

All bolts 60 mm long with exception of pole bolts.

Pin Angle – Additional round washers and locknuts required for bonding stay to pole top bolts.

+ For 33 kV applications, use 305 mm pins at all positions.



All dimensions are in millimetres unless otherwise stated

Drawing 434003 Single Pole, single crossarm, intermediate

Materials List for Drawing No. 434004
Intermediate or Pin Angle Pole (Double Crossarm Member)

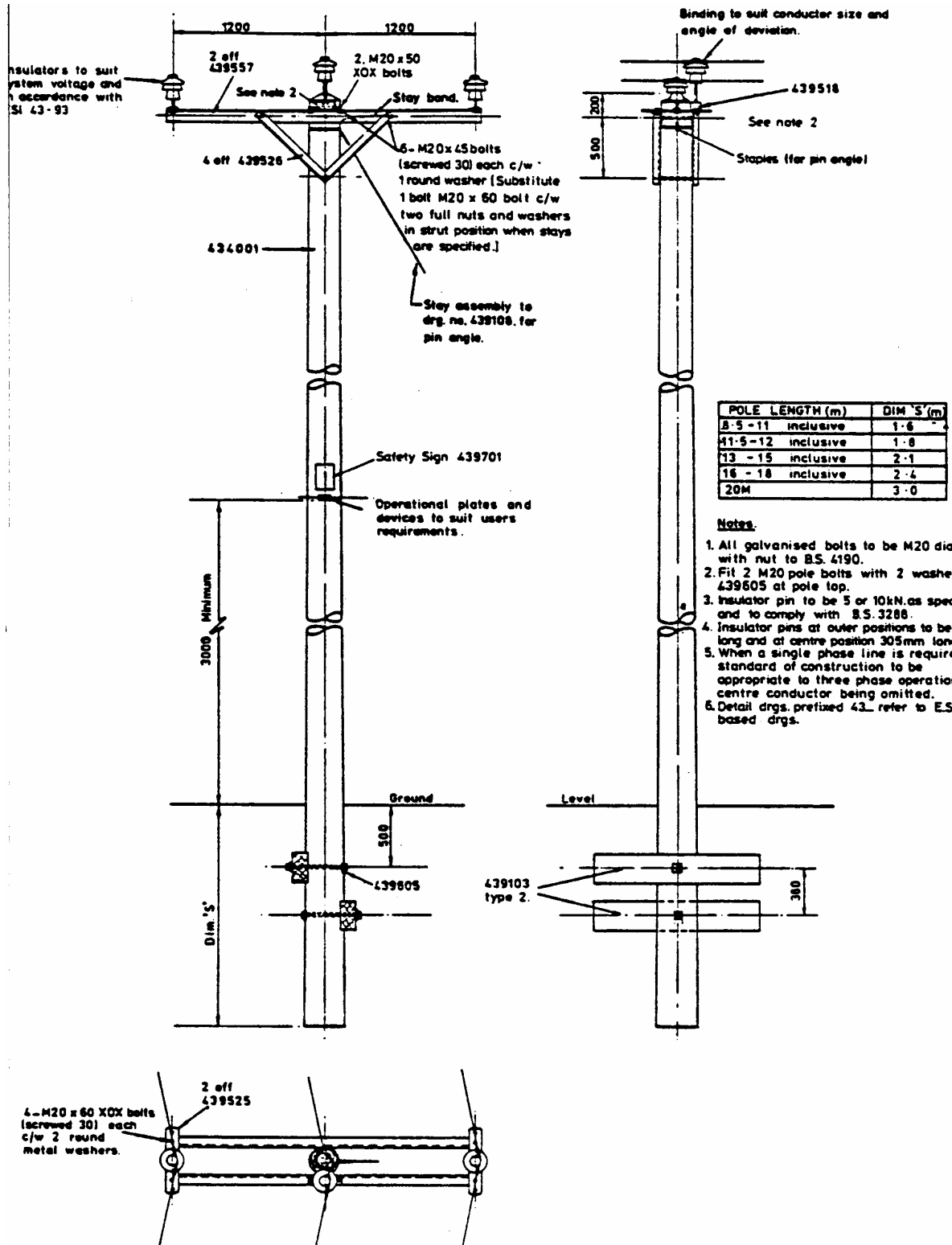
Quantity	Description	ENATS Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
1	Wood Pole	434001	4 (pole)	-
2	Kicking Block	439103 Type 2	2 (pole)	-
2	Crossarm Member Inter. & Angle/Section	439557	-	-
4	Crossarm Strut	439526	4	8
1	Insulator Bracket	439518	2	-
2	Section Strap	439525	4	8
2	Square Flat Washer	439604	-	-
6	Square Curved Washer	439605	-	-
3	11 kV Pin Insulator	439301	-	-
	20 kV Pin Insulator	439302	-	-
	33 kV Pin Insulator	439303	-	-
3	33 kV Line Post Insulator	439303	-	-
	11 kV Pin 230mm 5 kN	BS3288 Ref.28	-	- +
	11 kV Pin 230mm 10 kN	BS3288 Ref.29	-	-
	33 kV Pin 305mm 5 kN	BS3288 Ref.30	-	-
1	33 kV Pin 305mm 10 kN	BS3288 Ref.31	-	-
	Safety Sign	439004	-	-

Notes:

All bolts 60 mm long with exception of pole bolts.

Pin Angle – Additional round washers and locknuts required for bonding stay to pole top bolts

+ For 33 kV applications, use 305 mm pins at all positions.



All dimensions are in millimetres unless otherwise stated

Drawing 434004 Single Pole, double crossarm, intermediate

Materials List for Drawing No. 434005
Single Pole Through Straight Line Section or Section Angle

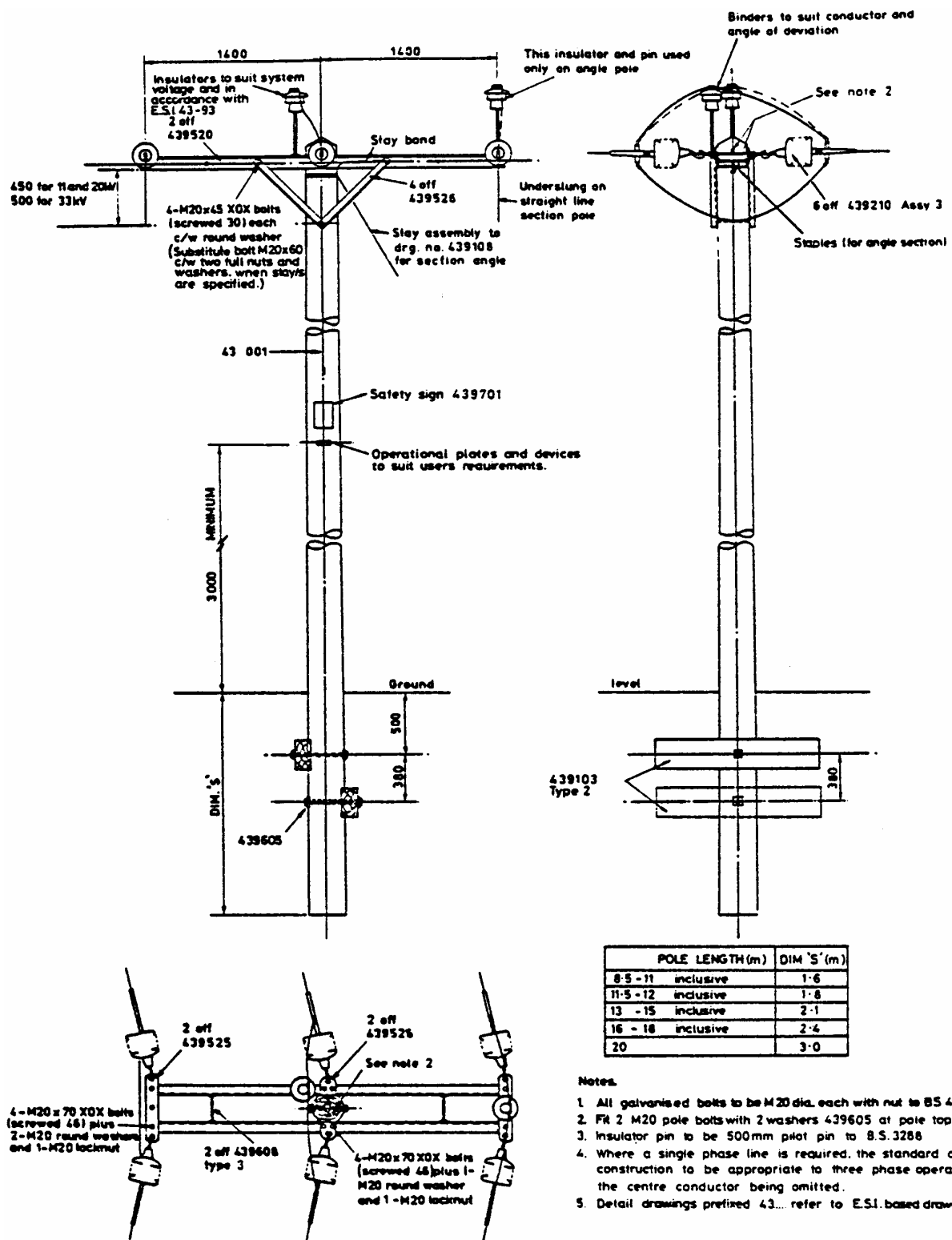
Quantity	Description	ENATS Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
1	Wood Pole	434001	4 (pole)	-
2	Kicking Block	439103 Type 2	2 (pole)	-
2	Crossarm Member	439520	-	-
4	Crossarm Strut	439526	4	- *
2	Section Strap	439525	4	4 +
2	Terminal Strap	439528	4	+
2	Square Flat Washer	439604	-	-
6	Square Curved washer	439605	-	-
1 or 2	11 kV Pin Insulator	439301	-	-
	20 kV Pin Insulator		-	-
	33 kV Pin Insulator	439302		
	33 kV Line Post Insulator	439303	-	-
1	Pilot Pin (50 mm Shank)	BS 3288 Ref. 50	-	-
6	Ball Ended Hook	BS 3288 Ref. 15/ 81	-	-
6	Insulator String	43 - 93	-	-
6	Conductor Termination	43 - 92	-	-
1	Safety Sign	439004	-	-

Notes:

All bolts 60 mm long with exception of pole bolts.

* Bolts to be complete with two full nuts and washers for bonding stay strand.

+ Lock nuts required.



All dimensions are in millimetres unless otherwise stated

Drawing 434005 Single Pole, double crossarm, section

Materials List for Drawing No. 434006
“H” Pole Straight Line Section or Section Angle

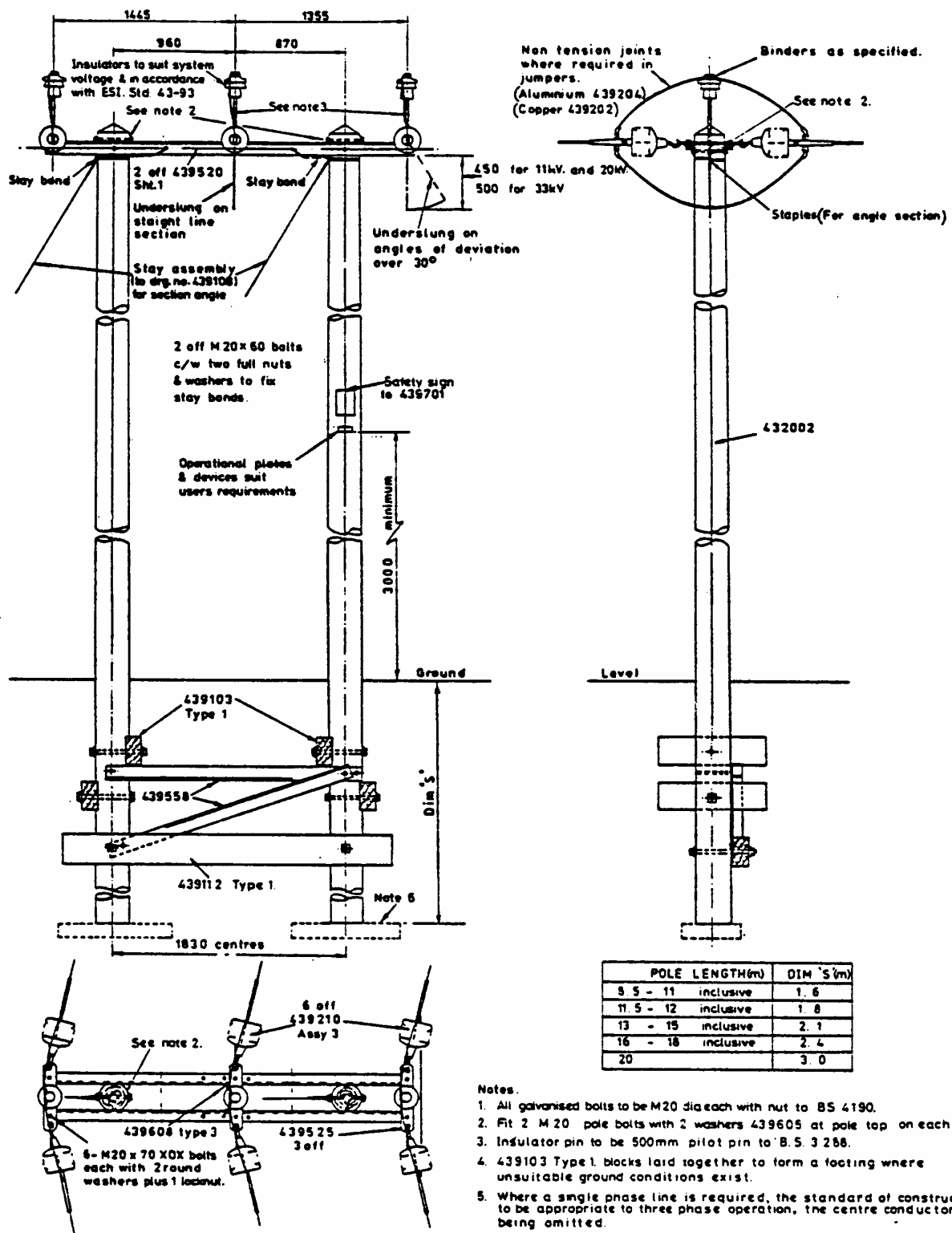
Quantity	Description	ENATS Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
2	Wood Pole	434002	6 (pole)	-
1	Brace Block	439112 Type 1	2 (pole)	-
2	Crossarm Member	439520	2	4 *
3	Section Strap	439525	6 (70mm)	12 +
6	Square Flat Washer	439604	-	-
16	Square Curved Washer	439605	-	-
2	Tie Rod	439608 Type 3	-	-
2	11 kV Pin Insulator	439301	-	-
	20 kV Pin Insulator		-	-
	33 kV Pin Insulator	439302	-	-
	33 kV Line Post Insulator	439303	-	-
2	Pilot Pin (50 mm Shank)	BS 3288 Ref. 50	-	-
6	Ball – ended Hook	BS 3288 Ref. 15/81	-	-
6	Insulator String	43 – 93	-	-
6	Conductor Termination	43 - 92	-	-
1	Safety Sign	439004	-	-
4	Wood Blocks	439103 Type 1	4 (pole)	-
2	Underground Brace Bars	439558	2 (pole)	-

Notes:

All bolts 60 mm long with exception of pole bolts and section strap (70 mm long).

* Bolts complete with two full nuts and washers only required at angle positions for stay bonding.

+ Lock Nuts required.



All dimensions are in millimetres unless otherwise stated

Drawing 434006 "H" Pole, double crossarm, section

Materials List for Drawing No. 434007

“H” Pole Straight Line Section or Section Angle for occasional long span

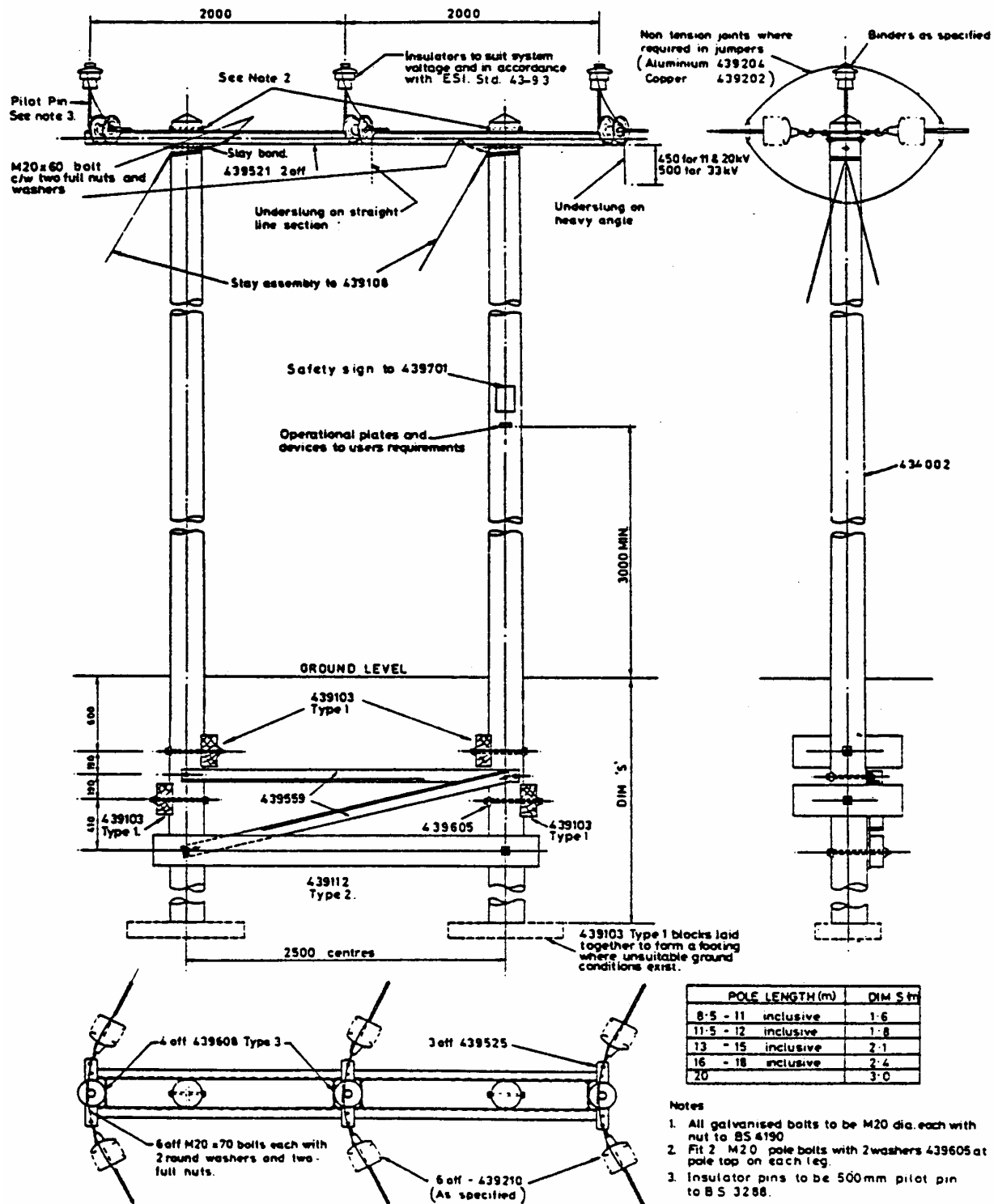
Quantity	Description	ENATS Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
1	Wood Pole	434002	6 (pole)	-
1	Brace Block	439112 Type 2	2 (pole)	-
2	Crossarm Member	439521	2	4 *
3	Section Strap	439525	6 (70mm)	12 +
6	Square Flat Washer	439604	-	-
16	Square Curved washer	439605	-	-
4	Tie Rod	439608 Type 3	-	-
2	11 KV Pin Insulator	439301	-	-
	20 KV Pin Insulator		-	-
	33 KV Pin Insulator	439302	-	-
	33 KV Line Post Insulator	439303	-	-
2	Pilot Pin (50 mm Shank)	BS 3288 Ref. 50	-	-
6	Ball – Ended Hook	BS3288 Ref.15/81	-	-
6	Insulator String	43 – 93	-	-
6	Conductor Termination	43 - 92	-	-
1	Safety Sign	439004	-	-
4	Wood Blocks	439103 Type 1	4 (pole)	-
2	Underground Tie Straps	439559	2 (pole)	-

Notes:

All bolts 60 mm long with exception of pole bolts and section strap (70 mm).

* Bolts complete with two full nuts and washers only required at angle positions for stay bonding.

+ Lock Nuts required.



Notes

1. All galvanised bolts to be M20 dia. each with nut to BS 4190
2. Fit 2 M20 pole bolts with 2 washers 439605 at pole top on each leg.
3. Insulator pins to be 500mm pilot pin to BS 3288.
4. Where a single phase line is required, the standard of construction to be appropriate to three phase operation, the centre conductor being omitted.
5. Detail drawings prefixed 43... refer to ESI based drawings

All dimensions are in millimetres unless otherwise stated

Drawing 434007 "H" Pole, double crossarm, occasional long span

**Materials List for Drawing No. 434008
Tee-Off Pole**

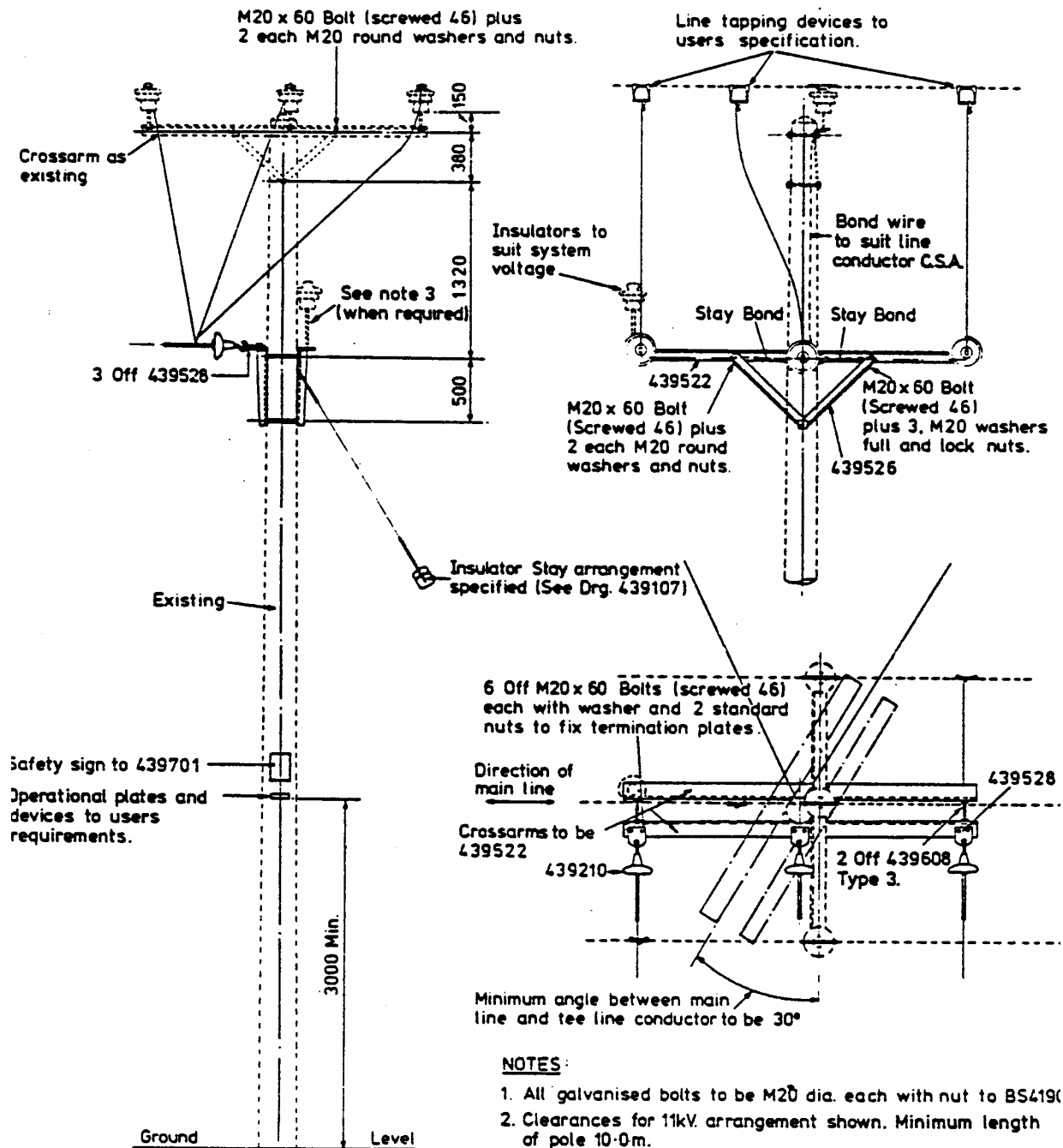
Quantity	Description	ENATS Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
2	Crossarm Member	439522	1 (pole)	2 *
4	Crossarm Strut	439526	2	4
2	Tie Rod	439608 Type 3	6	-
3	Terminal Strap	439528	6	12 +/-
3	Line Taps (as appropriate)		-	-
3	Ball Ended Hook	BS Ref 15/81	-	-
3	Insulator String	43 - 93	-	-
3	Conductor Termination	43 - 92	-	-
1	Safety Sign	439004	-	-
2	Square Curved Washer	439605	-	-

Notes:

All bolts 60 mm long with exception of pole bolts and terminal strap (70 mm).

* Bolts complete with two full nuts and washers only required at terminal positions for stay bonding.

+ Lock nuts required.



NOTES:

1. All galvanised bolts to be M20 dia. each with nut to BS4190
2. Clearances for 11kV. arrangement shown. Minimum length of pole 10.0m.
3. Insulator pin to be 500mm pilot in accordance with B.S. 3288.
4. When a single phase line is required, the standard of construction to be appropriate to three phase operation, the centre conductor being omitted.
5. Detail drawings prefixed 43..... refer to ESI. based drawings.

All dimensions are in millimetres unless otherwise stated

Drawing 434008 Single Pole, double crossarm, tee-off

Materials List for Drawing No. 434009
Single Pole Terminal (single member un-braced double crossarm)

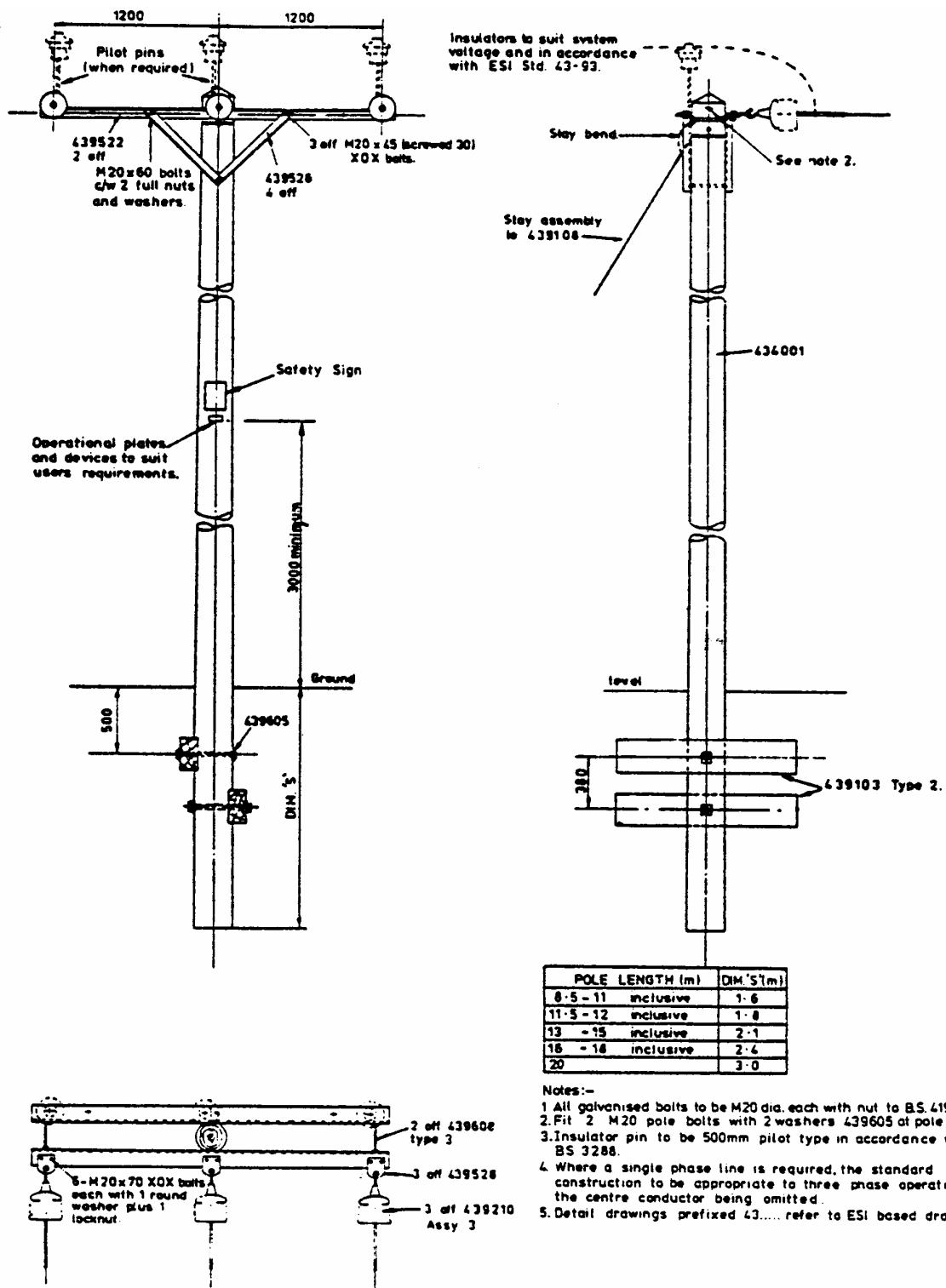
Quantity	Description	ENATS Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
1	Wood Pole	434001	3 (pole)	2
2	Kicking Block	439103 Type 2	2 (pole)	-
1	Crossarm Member	439522	-	-
2	Crossarm Strut	439526	4 *	-
2	Terminal Strap	439528	6 (70mm)	12 +
2	Square Flat Washer	439604	-	-
6	Square Curved washer	439605	-	-
2	Tie Rod	439608 Type 3	-	-
3	Ball Ended Hook	BS 3288 Ref. 15/81	-	-
3	Insulator String	43 – 93	-	-
3	Conductor Termination	43 - 92	-	-
1	Safety Sign	439004	-	-

Notes:

All bolts 60 mm long with exception of pole bolts and terminal strap (70 mm).

* Bolts complete with two full nuts and washers only required at angle positions for stay bonding.

+ Lock Nuts required.



All dimensions are in millimetres unless otherwise stated

Drawing 434009 Single Pole, un-braced double crossarm, terminal

Materials List for Drawing No. 434010
Single Pole Terminal (double member crossarm)

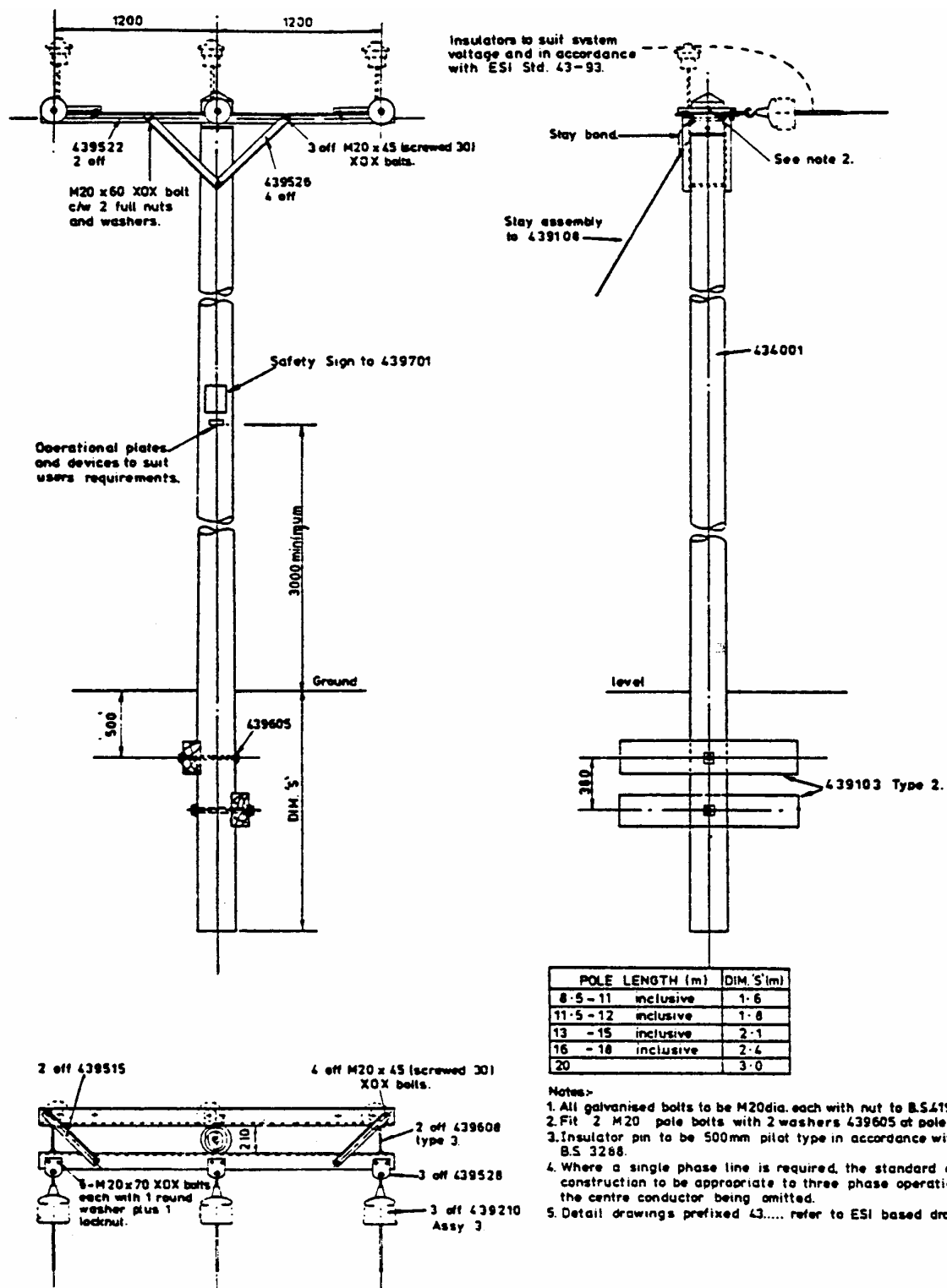
Quantity	Description	ENATS Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
1	Wood Pole	434001	3 (pole)	-
2	Kicking Block	439103 Type 2	2 (pole)	-
2	Crossarm Member	439522	-	-
4	Crossarm Strut	439526	3 *	6
1	Terminal Strap	439528	6 (70mm)	12 +
2	Crossarm Brace	439515	4	-
2	Square Flat Washer	439604	-	-
6	Square Curved washer	439605	-	-
2	Tie Rod	439608 Type 3	-	-
3	Ball – ended Hook	BS 3288 Ref. 15/81	-	-
3	Insulator String	43 – 93	-	-
3	Conductor Termination	43 - 92	-	-
1	Safety Sign	439004		-

Notes:

All bolts 60 mm long with exception of pole bolts and terminal strap (70 mm).

* Bolts complete with two full nuts and washers only required at terminal positions for stay bonding.

+ Lock Nuts required



All dimensions are in millimetres unless otherwise stated

Drawing 434010 Single Pole, double crossarm, terminal

**Materials List for Drawing No. 434011
"H" Pole Terminal**

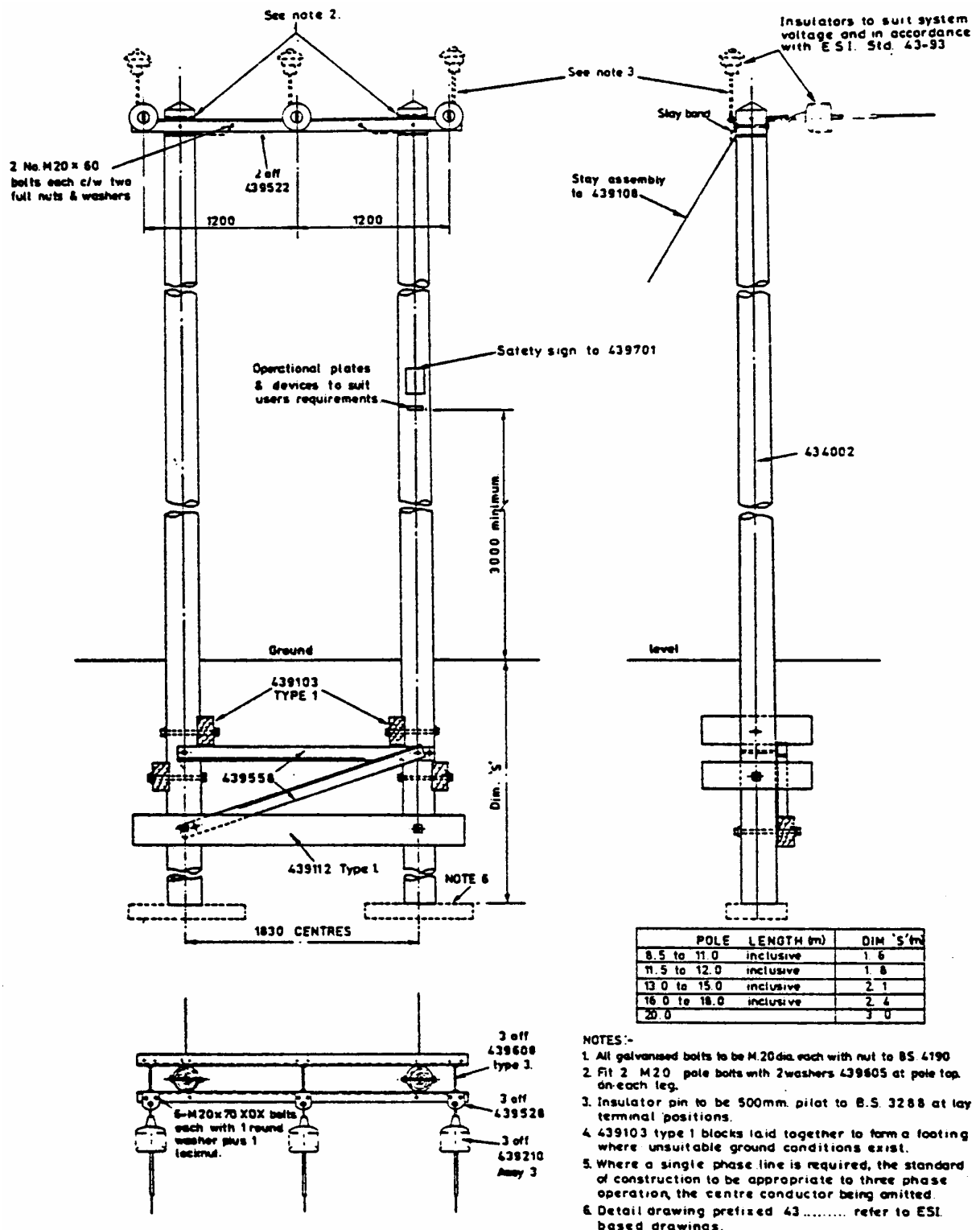
Quantity	Description	ENATS Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
1	Wood Pole	434002	6 (pole)	
1	Brace Block	439112 Type 1	2 (pole)	
2	Crossarm Member	439522	2 *	4
3	Terminal Strap	439528	6	12 +
6	Square Flat Washer	439604	-	-
8	Square Curved Washer	439605	-	-
3	Tie Rod	439608 Type 3	-	-
3	Ball Ended Hook	BS 3288 Ref 15/ 81	-	-
3	Insulator String	43 - 93	-	-
3	Conductor Termination	43 - 92	-	-
1	Safety Sign	439004	-	-
2	Underground Tie Straps	439558	2 (pole)	
4	Wood Blocks	439103 Type 1	4 (pole)	

Notes:

All bolts 60 mm long with exception of pole bolts and terminal strap (70 mm).

* Bolts complete with two full nuts and washers only required at terminal positions for stay bonding.

+ Lock Nuts required



All dimensions are in millimetres unless otherwise stated

Drawing 434011 "H" Pole, double crossarm, terminal

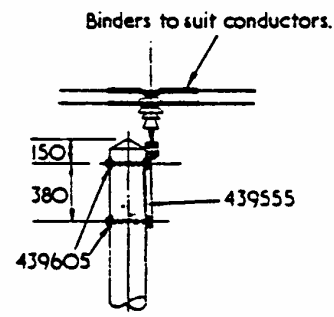
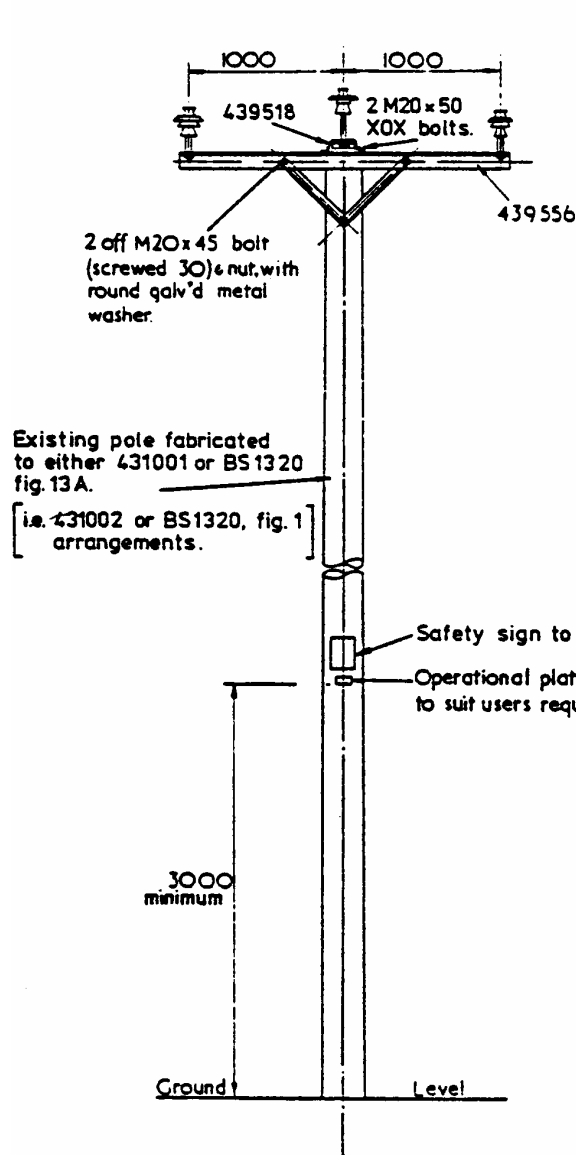
**Materials List for Drawing No. 434012
Intermediate Pole Top Renovation**

Quantity	Description	ENATS Dwg. No. or BS Ref.	M 20 Bolts	Round Washers
1	Crossarm Member	439556		2
2	Crossarm Strut	439555	4	10
1	Insulator Bracket	439518	6	12
2	Square Curved Washer	439605	-	-
2	11 kV Pin 230 mm 5 kN	BS Ref. 28	-	- +
1	11 kV Pin 230 mm 10 kN	BS Ref. 29		
	33 kV Pin 305 mm 5 kN	BS Ref. 30		
	33 kV Pin 305 mm 10 kN	BS Ref. 31		
3	11 kV Pin Insulator	439301	-	-
	20 kV Pin Insulator			
	33 kV Pin Insulator	439302		
	33 kV Line Post Insulator	439303		
1	Safety Sign	439004	-	-

Notes:

All bolts 60 mm long with exception of pole bolts.

+ For 33 kV application, use 305 mm pins at all positions.



NOTES.

1. All galvanised bolts to be M20 dia. each with nut to BS4190.
2. Insulator pin to be 5 or 10kN as specified and to comply with BS 3288.
3. Insulator pins at outer positions to be 230 long and at centre position 305 long.
4. Where a single phase line is required, the standard of construction to be appropriate to three phase operation, the centre conductor being omitted.
5. Detail drawings prefixed 43..... refer to ESI based drawings.

All dimensions are in millimetres unless otherwise stated

Drawing 434012 Single Pole, single crossarm, renovation

APPENDIX A

USER GUIDE TO MICROSOFT EXCEL WORKBOOK 4340_EMPIRICAL.XLS

A1 INTRODUCTION

Historically, overhead line specifications such as ENATS 43-10 (11 kV Single Circuit Overhead Lines of Light Construction on Wood Poles) and ENATS 43-20 (11 kV and 33 kV Single Circuit Overhead Lines of Heavy Construction on Wood Poles) have generally provided information to the design engineer that eliminates the need for complicated calculations. This was achieved by providing lookup tables which covered a number of “standard” conductors along with typical span limits and addressed many of the situations encountered in the field.

In view of the sheer range of conductors now in use by distribution network operators and the differences in weather loadings for locality and altitude, it would be impractical to provide in tabulated form, component suitability relating to each of the conductors in use. A series of spreadsheets comprising a workbook has therefore been developed, in Microsoft Excel format, which allows the User the flexibility to choose conductor systems specific to his system requirements. Users may now determine maximum allowable windspan, angles of deviation, gradients, strut loading, foundation and stay suitability for each structure, by the application of User specified weather conditions and User selected factors of safety. The results from each spreadsheet may then be compiled and tabulated to provide the User with their own specific line design.

This User Guide provides a description of the Microsoft Excel workbook spreadsheets (section A2) and a worked example with tabulated results (section A3, with Tables appended).

A2 SPREADSHEET DESCRIPTIONS

A2.1 README 1ST

This initial page provides the User with a general introduction to the spreadsheets contained in the package. Rules are set for working through the document in relation to the spreadsheets used. The User is encouraged, in addition to referencing these guidance notes, to familiarise themselves with the contents of the README 1ST sheet prior to carrying out work associated with the Workbook.

A2.2 DESIGN PARAMETERS

This spreadsheet, as Figure 1 below indicates, requires basic line design parameters to be used in all subsequent spreadsheets. The factors of safety can be altered at the Users discretion. The unstayed pole factor of safety should be increased as the downpull increases as detailed in 6.3 of this Specification.

The maximum wind span length required is different to the maximum individual span length, and is calculated as half the maximum sum of adjacent spans. *The User should ensure that the maximum wind span value entered on this spreadsheet does not exceed that included in the crossarm design spreadsheets SINGLE POLE FIXING and H POLE FIXING.*

When necessary, Users may additionally include the weight of pole-mounted equipment to be included in the strut loading on a pole.

Line Voltage (kV)	11
Number of Conductors	3
Maximum Wind Span (m)	105
<i>FACTORS of SAFETY:</i>	
Unstayed Pole	3.5
Foundation	2.5
Stayed Pole	2.5
Stays	2.5
Steelwork & Bolts (UTS)	2.5
Pin Insulator	2.5
Equipment Weight (kgf)	112

FIGURE 1

A2.3 SAGNTEN

This spreadsheet is used to process the relevant data relating to the conductor considered for design.

There are three parts to this spreadsheet that allow the User to compile information relating to the mechanical performance of the conductor selected and data for preparation of a sag template.

A2.3.1 Conductor Sag/ Tension Data for Design

The information contained in this section provides all relevant information for the chosen conductor and is used to calculate values relating to the conductor's tension and sag over a predetermined basic / recommended span (see Figure 2). The maximum conductor working tension, weight, wind pressure, and freezing point tension values are all used by other spreadsheets to determine component suitability.

Conductor Code Name (if any)	OAK
Greased Conductor Weight (kg/m)	0.3273
Cross Sectional Area of Conductor (mm ²)	118.9
Conductor Overall Diameter (mm)	14
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm ²)	6016.33
Rated Breaking Strength of Conductor (kgf)	3576.1
Basic / Recommended Span (m)	110
Wind Pressure on Conductor (N/m ²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1788.1
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	715.2
Temperature at EDT Limit (Degrees C)	5
Maximum Conductor Tension (MCT) (kgf) at	-5.6°C
Maximum Conductor Weight (MCW) (kg/m)	0.968
Maximum Conductor Pressure (MCP) (kg/m)	1.279
Freezing Point Tension (FPT) (kgf) at	0°C
	784.7

FIGURE 2

A2.3.2 Ordinates for Plotting Uplift & Ground Clearance Templates

As the title suggests, this section of the spreadsheet is used to produce data to enable construction of a plastic sag template that could then be applied to overhead line profiles. The User is required to convert the span and sag data to the scales as employed on the profile sheet. A temperature shift facility has been provided, although this is not normal practice and advice should be sought if this is to be considered.

Conductor: OAK

Temperature Shift (Deg. C)

0

Tension Reduction at 15°C (%)

0

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.08	0.21	0.23	0.24	0.26	0.27	0.29
80	0.30	0.85	0.91	0.97	1.03	1.09	1.15
120	0.68	1.92	2.05	2.19	2.32	2.45	2.58
160	1.21	3.40	3.65	3.89	4.13	4.36	4.59
200	1.89	5.32	5.71	6.08	6.45	6.81	7.16
240	2.72	7.66	8.22	8.76	9.29	9.81	10.32
280	3.71	10.43	11.18	11.93	12.65	13.36	14.04
320	4.84	13.62	14.61	15.58	16.52	17.44	18.34
360	6.13	17.24	18.49	19.72	20.91	22.08	23.21
400	7.56	21.28	22.82	24.34	25.82	27.26	28.66
440	9.15	25.75	27.62	29.45	31.24	32.98	34.67
480	10.89	30.64	32.87	35.05	37.18	39.25	41.27
520	12.78	35.96	38.57	41.13	43.63	46.07	48.43

FIGURE 3

A2.3.3 Design & Erection Tables

The information shown on this part of the spreadsheet (see Figure 4) provides the User with the data necessary to carry out design profiling and also data relating to on-site construction and erection of the conductor. In order to obtain data for erection of the conductor, a temperature shift or percentage increase can be input to the specified fields. Whenever this action is carried out the heading will automatically change to "ERECTION TABLE".

Conductor Code Name	OAK
Basic / Recommended Span (m)	110
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	865.7	0.17	0.23	0.30	0.38	0.47	0.57	0.68	0.80	0.93
0	784.7	0.19	0.26	0.33	0.42	0.52	0.63	0.75	0.88	1.02
5	715.2	0.21	0.28	0.37	0.46	0.57	0.69	0.82	0.97	1.12
10	649.1	0.23	0.31	0.40	0.51	0.63	0.76	0.91	1.07	1.24
15	587.2	0.25	0.34	0.45	0.56	0.70	0.84	1.00	1.18	1.37
20	530.3	0.28	0.38	0.49	0.62	0.77	0.93	1.11	1.30	1.51
25	479.0	0.31	0.42	0.55	0.69	0.85	1.03	1.23	1.44	1.67
30	433.7	0.34	0.46	0.60	0.76	0.94	1.14	1.36	1.59	1.85
35	394.4	0.37	0.51	0.66	0.84	1.04	1.26	1.49	1.75	2.03
40	360.7	0.41	0.56	0.73	0.92	1.13	1.37	1.63	1.92	2.22
45	332.0	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	2.42
50	307.6	0.48	0.65	0.85	1.08	1.33	1.61	1.92	2.25	2.61
55	286.8	0.51	0.70	0.91	1.16	1.43	1.73	2.05	2.41	2.80
60	268.9	0.55	0.75	0.97	1.23	1.52	1.84	2.19	2.57	2.98
65	253.5	0.58	0.79	1.03	1.31	1.61	1.95	2.32	2.73	3.16
70	240.2	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.88	3.34
75	228.4	0.64	0.88	1.15	1.45	1.79	2.17	2.58	3.03	3.51
80	218.1	0.68	0.92	1.20	1.52	1.88	2.27	2.70	3.17	3.68

FIGURE 4

A2.4 COND SPACING

This spreadsheet permits Users to determine the maximum span before conductor clash is likely to occur and is related to the “weather zone” for the specific locality and elevation. It is recommended that a “2B” weather zone be employed as a minimum for “*Normal*” altitudes, whilst for “*High*” altitudes a weather zone of “3C” be employed as a minimum. More onerous weather zones may be employed as indicated on the weather mapping included in Section 8 of this Specification. Allowance has been included for lines where adjacent conductors are not horizontal (ie. centre phase stool). When considering angle arrangements, Users should allow for the effect of the line angle when assessing spacing. For example, a 60° heavy angle crossarm will reduce the standard 439520 crossarm spacing to $1400 \times \cos(60 / 2)^\circ = 1212 \text{ mm}$.

Conductor Code Name (if any)	OAK
Weather Index (from ENATS 43-40 maps)	2B
Basic / Recommended Span (m)	110
Freezing Point Tension (kgf)	784.7
Horizontal Conductor Offset or Spacing (mm)	1200
Vertical Conductor Offset or Spacing (mm)	0
Unit Load Reliability Factor	1.00
Spacing Withstand Factor	1.10
Maximum Span to Avoid Clash (m)	124.16
Clash Factor	1.129

FIGURE 5

Once all spacing / weather zones data has been entered on the spreadsheet, it is essential that the “Maximum Span to Avoid Clash” button be pressed, as reminded by the pop-up comment.

A2.5 S STRUT & H STRUT

These spreadsheets show strut loadings imposed on the structures as a result of the stay combination, angle of deviation and the angle of stay slope. The gradient of the line is also used to determine the ultimate strut loadings on the poles. Normally a maximum gradient will be considered for structure types and these are usually in the order of 1 : 10 (5.7°) or 1 : 5 (11.3°). The strengths produced in the spreadsheet must then be the limiting values when cross referencing the capability of the wood pole in the WOODPOLE spreadsheet, i.e. when choosing a pole for a specific location, the maximum permissible strut loading in WOODPOLE must not be exceeded by the value produced in the appropriate cell referenced in the S or H STRUT spreadsheet (Figure 6). The figure below is an example of the S STRUT spreadsheet.

Conductor Code Name (if any)			OAK	
Maximum Conductor Tension (MCT) (kgf)			1334.7	
Wind Pressure (N/m ²)			380	
Maximum Conductor Weight (MCW) (kg/m)			0.968	
Maximum Conductor Pressure (MCP) (kg/m)			1.279	
Maximum Wind Span Length (m)			115.0	
Number of Conductors			3	
Factor of Safety			2.5	
Wind Assumed on 18m STOUT Pole			y	
Longitudinal Load in Conductors (kgf)			10010.3	
Transverse Load [Normal to Span] (kgf)			1102.9	
Vertical Load [No Downpull] (kgf)			1662.9	
Vertical Load with Downpull 1:	10	(kgf)	3655.0	5.71 Degrees Declin.
Pole Wind Loading (kgf)			224.5	

STRUT LOAD in POLE with ONE or TWO STAYS (kgf)				
Line Angle (Degrees)	Stay Angle (Degrees)			
	30	35	40	45
Level Conditions				
0	3962	3559	3245	2990
5	5473	4804	4284	3863
10	6977	6045	5319	4731
15	8472	7277	6348	5594
20	9955	8500	7368	6450
25	11422	9710	8378	7297
30	12872	10905	9375	8134
35	14301	12084	10359	8960
40	15707	13243	11326	9771
45	17087	14381	12275	10568
50	18438	15495	13205	11348
55	19758	16583	14113	12110
60	21044	17644	14999	12853
Downpull 1: 10				
0	5954	5551	5237	4982
5	7465	6796	6276	5855
10	8969	8037	7311	6723
15	10464	9269	8340	7586
20	11947	10492	9360	8442
25	13414	11702	10370	9290
30	14864	12897	11367	10127
35	16293	14076	12351	10952
40	17699	15235	13318	11763
45	19079	16373	14268	12560
50	20430	17487	15197	13340
55	21750	18575	16106	14102
60	23037	19636	16991	14845

FIGURE 6

A2.6 WOODPOLE

This spreadsheet is used to indicate three specific issues relating to the wood pole capability:

- Maximum strut loading permissible in relation to the given pole dimensions.
- Maximum permissible windspan the pole is capable of achieving prior to failure.
- The maximum windspan capability relative to the foundation of the pole, aided or unaided with one, two or no blocks attached to the pole in either Good, Good to Average or Average to Poor soil conditions.

Design Fibre Stress of Pole and Blocks (N/mm ²)	53.3	
Design Average Modulus of Elasticity (N/mm ²)	10054	
Wind Pressure (N/m ²)	380	
Pole Factor of Safety	2.5	
Foundation Factor of Safety	2.5	
Number of Conductors	3	
Maximum Conductor Pressure (MCP) (kg/m)	1.279	
Wind on Pole Considered	y	
Average Conductor Level Above Pole Top (mm)	160	
Number of Blocks (0, 1[Upper], 2)	0	
Top Block Length (mm) (Normal Section 250 x 125mm)	0	
Groundline to Top Block Pole Bolt (mm)	0	
Bottom Block Length (mm) (Normal Section 250 x 125mm)	0	
Groundline to Bottom Block Pole Bolt (mm)	0	
Soil Type (Good, Good/Average, Average/Poor)	ga	(Enter G, GA, or AP)

POLE DATA TABLE							
Length (m)	Grade	Top Diam. (mm)	Dia. 1.5m from Butt (mm)	Planting Depth (mm)	Single Pole Strut Strength (kgf)	Maximum Wind Span Length for Specified Pole (m)	Maximum Wind Span Length for Specified Found. (See Key Below)
8.5	MEDIUM	150	215	1600	7768	68	>>>>>> 39
9	MEDIUM	150	220	1600	7018	68	>>>>>> 37
9.5	MEDIUM	150	225	1600	6395	67	>>>>>> 35
10	MEDIUM	150	230	1600	5874	68	>>>>>> 33
10.5	MEDIUM	150	235	1600	5431	68	>>>>>> 31
11	MEDIUM	150	240	1600	5051	68	>>>>>> 29
11.5	MEDIUM	150	245	1800	4853	68	>>>>>> 44
12	MEDIUM	150	250	1800	4550	69	>>>>>> 42
13	MEDIUM	160	260	2100	4814	70	>>>>>> 69
14	MEDIUM	160	275	2100	4476	75	>>>>>> 66
15	MEDIUM	165	290	2100	4488	81	>>>>>> 62
16	MEDIUM	170	305	2400	4645	87	<<<87
17	MEDIUM	180	320	2400	4979	94	>>>>>> 94
18	MEDIUM	180	330	2400	4623	95	>>>>>> 89
20	MEDIUM	180	360	3000	4468	103	<<<104
22	MEDIUM	190	380	3000	4455	109	<<<109
8.5	STOUT	190	265	1600	19025	134	>>>>>> 48
9	STOUT	190	275	1600	17629	140	>>>>>> 46
9.5	STOUT	190	280	1600	15932	138	>>>>>> 43
10	STOUT	190	285	1600	14513	136	>>>>>> 40
10.5	STOUT	190	290	1600	13313	135	>>>>>> 38
11	STOUT	190	295	1600	12286	134	>>>>>> 36
11.5	STOUT	190	300	1800	11725	133	>>>>>> 54
12	STOUT	190	305	1800	10915	133	>>>>>> 51
13	STOUT	195	320	2100	10815	139	>>>>>> 85
14	STOUT	195	335	2100	9866	144	>>>>>> 80
15	STOUT	195	350	2100	9113	148	>>>>>> 75
16	STOUT	200	365	2400	9192	154	>>>>>> 117
17	STOUT	200	375	2400	8402	153	>>>>>> 110
18	STOUT	200	390	2400	7946	156	>>>>>> 106
20	STOUT	200	415	3000	7323	158	<<<158
22	STOUT	200	435	3000	6468	155	<<<156
24	STOUT	200	470	3000	6238	164	<<<164

FIGURE 7

The column detailing the 'Maximum Wind Span Length for Specified Found', shown in Figure 7, automatically defaults to the lower value of windspan when comparing pole and foundation capabilities. This span length is then used as the maximum windspan permissible for the specific conductor/ span/ foundation arrangement.

A2.7 STRUT FOUND

This spreadsheet provides the User with allowable soil-bearing loads with the specified pole (from the WOODPOLE spreadsheet) and a defined foundation blocking arrangement, based on an average soil bearing capacity of 429 kN/m² (4 tons/ft²). The weight of poles considered in the spreadsheet is based on 1000 kgf for MEDIUM grade poles and 1600 kgf for STOUT grade poles.

Top Block Length (mm) (Normal Section 250 x 125mm)	1300
Bottom Block Length (mm) (Normal Section 250 x 125mm)	1300
Cross Baulk Length (mm) for "H" Pole	2600
Cross Baulk Width (mm) for "H" Pole	125
Average Soil Bearing Resistance (kN/ m ²)	429

POLE TABLE and ALLOWABLE STRUT LOADS					
Pole Length (m)	Pole Grade	Top Diam. (mm)	Dia. 1.5m from Butt (mm)	Maximum Allowable Load in Single Pole (kgf)	Maximum Allowable Load in "H" Pole (kgf)
8.5	MEDIUM	150	215	15018	41519
9	MEDIUM	150	220	15099	41681
9.5	MEDIUM	150	225	15181	41845
10	MEDIUM	150	230	15265	42013
10.5	MEDIUM	150	235	15350	42184
11	MEDIUM	150	240	15438	42359
11.5	MEDIUM	150	245	15527	42536
12	MEDIUM	150	250	15617	42718
13	MEDIUM	160	260	15779	43041
14	MEDIUM	160	275	16083	43649
15	MEDIUM	165	290	16390	44264
16	MEDIUM	170	305	16713	44909
17	MEDIUM	180	320	17040	45563
18	MEDIUM	180	330	17275	46032
20	MEDIUM	180	360	18039	47560
22	MEDIUM	190	380	18548	48580
8.5	STOUT	190	265	15332	42147
9	STOUT	190	275	15547	42577
9.5	STOUT	190	280	15646	42774
10	STOUT	190	285	15746	42975
10.5	STOUT	190	290	15849	43180
11	STOUT	190	295	15953	43389
11.5	STOUT	190	300	16059	43601
12	STOUT	190	305	16167	43818
13	STOUT	195	320	16503	44490
14	STOUT	195	335	16870	45223
15	STOUT	195	350	17251	45985
16	STOUT	200	365	17633	46749
17	STOUT	200	375	17895	47274
18	STOUT	200	390	18316	48116
20	STOUT	200	415	19042	49568
22	STOUT	200	435	19643	50769
24	STOUT	200	470	20800	53082

FIGURE 8

A2.8 S STAYS & H STAYS

These spreadsheets provide the User with the maximum permissible angle of line deviation based on the angle of stay slope, stay type, number of stays used, and the loads imposed on these stays from the conductor loading regime used. The Figure below shows an example of the S STAYS spreadsheet.

Conductor Code Name (if any)	OAK
Maximum Conductor Tension (MCT) (kgf)	1334.7
Wind Pressure (N/m ²)	380
Maximum Conductor Pressure (MCP) (kg/m)	1.279
Maximum Wind Span Length (m)	115
Number of Conductors	3
Factor of Safety	2.5
Wind Assumed on 18m STOUT Pole	y

Where two stays used, calculation based on 2m separation between stays at ground level.

Angle of Stay Slope	MAXIMUM ANGLE of LINE DEVIATION								User Defined Stay(s) No. 1 6281
	Grade 700 Staywire				Grade 1150 Staywire				
	1x7/4.00	2 x 7/4.00	1x19/3.55	2x19/3.55	1x7/3.25	2 x 7/3.25	1x7/4.00	2x7/4.00	
	Stay Strength (kgf)				Stay Strength (kgf)				
	6281	6281	11216	11216	6628	6628	10299	10299	
30	10	27	24	57	11	29	22	51	10
35	13	33	29	60	14	35	26	60	13
40	15	38	34	60	16	41	30	60	15
45	17	44	38	60	19	47	34	60	17

FIGURE 9

Single pole structures are not always capable of carrying the full mechanical loads required. It is therefore necessary to consider introducing a stronger structure such as an H pole in such circumstances. The S STAY and H STAY spreadsheets also provide approximate stay spreads relative to the angle of stay slope as shown in the following Figure 10.

Overall Pole Length (m) 10	Pole Planting Depth (m m) 1600	Stay Position From Pole Top (m m) 175
Angle of Stay Slope		Stay Spread (m m)
25		3835
26		4012
27		4191
28		4373
29		4559
30		4749
31		4942
32		5140
33		5341
34		5548
35		5759
36		5976
37		6198
38		6426
39		6660
40		6902
41		7150
42		7406
43		7670
44		7943
45		8225
46		8517
47		8820
48		9135
49		9462
50		9802

FIGURE 10

A2.9 STAY FOUND

This spreadsheet indicates whether the full stay tensile load may be achieved by the soil frustum, and also the minimum stay separation distances for totally independent stay foundations. The User may vary stay parameters as necessary.

Block Length (mm)	1300
Block Width (mm)	250
Block Depth (mm)	125
Stay Rod Length (mm)	2500
Maximum Stay Strength (kgf)	10299
Soil Density [from ENATS 43-91] (kg/m ³)	1595
Soil Frustum Angle (Degrees)	30

Stay Angle	30	35	40	45
Volume of Foundation (m ³)	8.3	7.2	6.1	5.0
Soil Uplift Weight (kgf)	13199.0	11493.2	9739.1	8003.9
Allowable Stay Tension (kgf)	15241	14031	12714	11319
FOUNDATION ADEQUACY	##OK!##	##OK!##	##OK!##	##OK!##
Recommended Stay Separation (m)	3.5	3.3	3.2	3.1

FIGURE 11

A2.10 CROSSARM MEMBER & CROSSARM STRUT

The steelwork section dimensions used will need to be varied whenever differing section size / bolt spacing dimensions are introduced. For crossarms designed to ENATS 43-95, the correct dimensions can be found by making reference to the appropriate crossarm drawing within that Specification, and entering these into the relevant cells on this spreadsheet. FIGURE 12 for example, shows the dimensions for a Terminal crossarm (ENATS 439522). The hole in the horizontal flange will only need to be included when used in single pole terminal crossarm arrangements.

Similarly, in the CROSSARM STRUT spreadsheet, the strut dimensions may be varied where these differ from those indicated on drawing ENATS 439526.

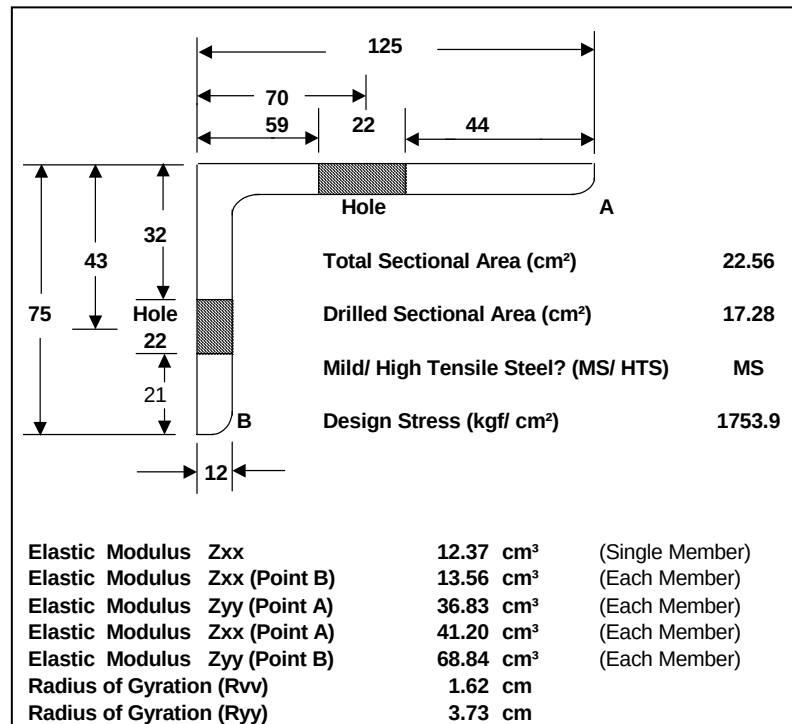


FIGURE 12

A2.11 SINGLE POLE FIXING

This spreadsheet provides the User with information relating to the ability of the crossarm to withstand loads relative to the conductor being used. It is important that any information entered into the blue fields of this spreadsheet relate directly to the specific crossarm being used. Should an alternative crossarm steel member section be considered, the User should additionally amend the appropriate cells in the CROSSARM MEMBER spreadsheet also.

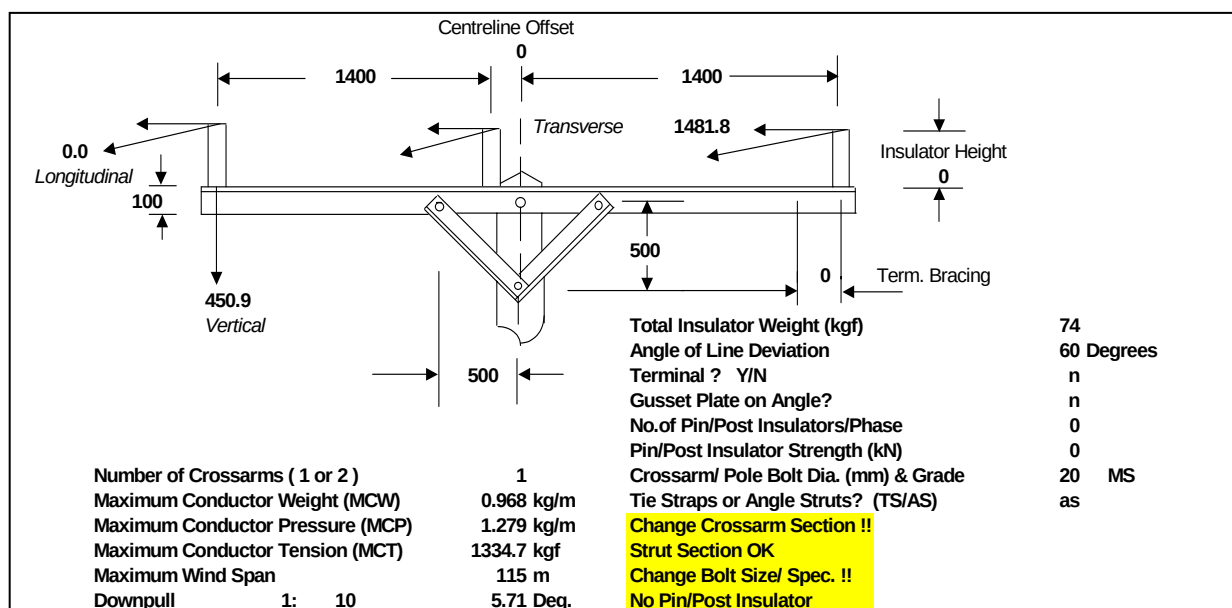


FIGURE 13

The maximum wind span cell may be amended as necessary, however it should not exceed the span length included in the DESIGN PARAMETERS spreadsheet.

A2.12 H POLE FIXING

The User should enter data relative to the “H” Pole crossarm hamper in the same way as the previous spreadsheet, SINGLE POLE FIXING, being careful to incorporate the correct pole / phase spacing. Figure 14 below shows the correct spacing for the standard H pole angle / section structure (drg. ENATS 434006).

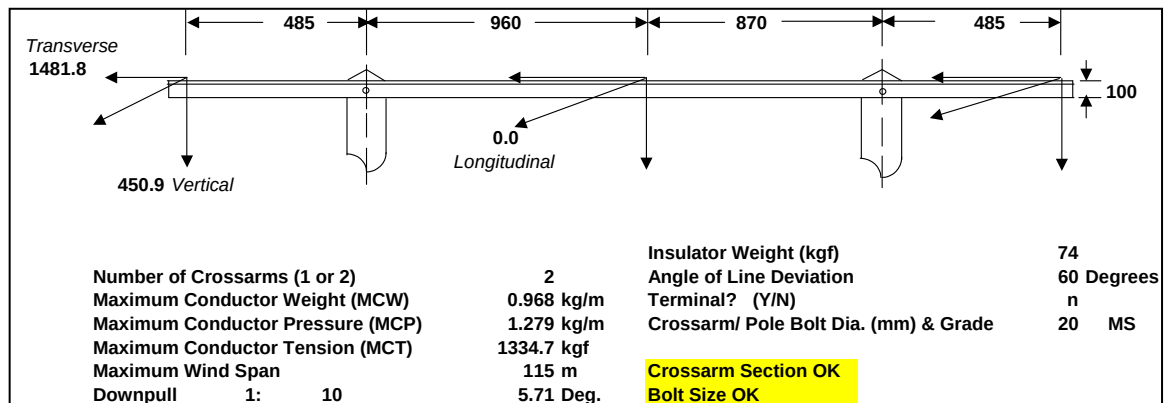


FIGURE 14

A2.13 TERM STRAP

This spreadsheet considers the applied loadings to the line and terminal straps used to land the tension insulator strings. The calculation takes into account the position and number of fixing holes and/ or slots in the plate. Note that the maximum full span length has been used in this calculation.

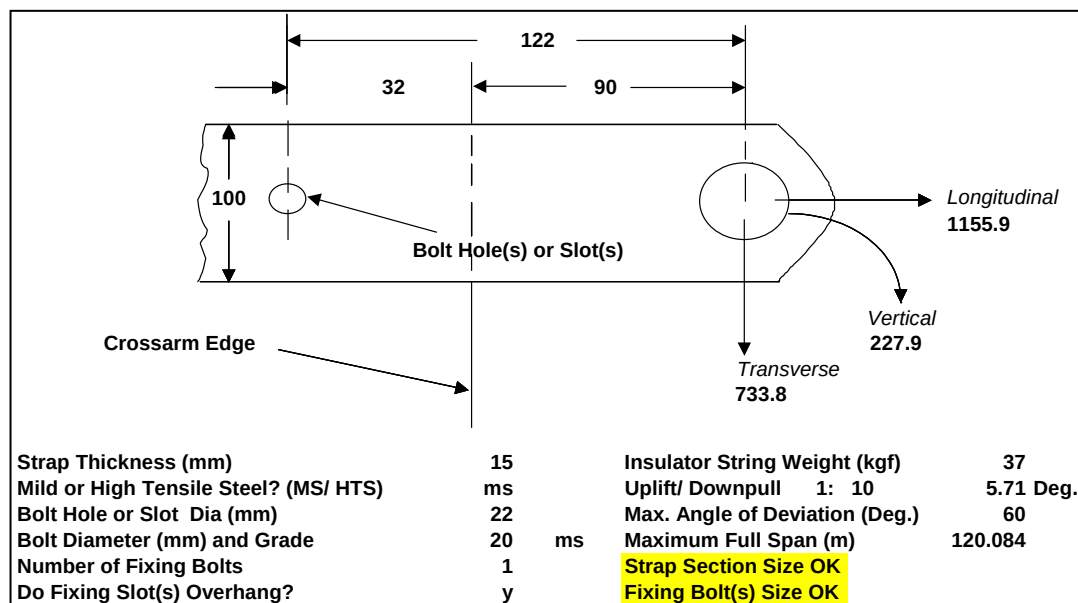


FIGURE 15

A2.14 COND DATABASE

As the title suggests, this spreadsheet contains all of the relevant data for each of a number of “standard” conductors, although the User may freely amend any of the parameters included, or add new conductors as required. The data is stored here for reference and is recalled by the User when completing the SAGNTEN spreadsheet for the specific conductor chosen.

Conductor parameters have been taken from BS EN 50182 (for aluminium-based conductors) and BS 7884 (for copper-based conductors). *Note that the conductor parameters in BS EN 50182 and BS 7884 vary slightly from those included in the now withdrawn standards BS 3242, 215 and 125.*

It is important when entering the conductor weight that allowance is made for the grease weight. In all cases, bare conductor weights have been augmented by the minimum grease weights in ENAER L38/ 1. Where higher greasing levels are required in particularly aggressive environments (ie. Category 4 for larger conductors), Users will need to amend conductor weights accordingly.

A vibration limit of 20 % of rated breaking strength has been assumed for all aluminium-based conductors, and 33 % for all copper-based conductors. It is not advisable that these vibration limits be amended.

To enable the User to rapidly copy the relevant conductor to the SAGNTEN spreadsheet, a number of macros have been developed, and the macro command codes are as detailed in the following table:

Macro Command	Old Code Name	New BSEN 50182 Code
Ctrl + a	150mm ² AAAC "ASH"	181-AL3
Ctrl + d	100mm ² ACSR "DOG"	105-AL1/14-ST1A
Ctrl + e	175mm ² AAAC "ELM"	211-AL3
Ctrl + f	40mm ² AAAC "FIR"	48-AL3
Ctrl + g	25mm ² ACSR "GOPHER"	26-AL1/4-ST1A
Ctrl + h	50mm ² AAAC "HAZEL"	60-AL3
Ctrl + j	200mm ² ACSR "JAGUAR"	211-AL1/12-ST1A
Ctrl + l	175mm ² ACSR "LYNX"	183-AL1/43-ST1A
Ctrl + m	25mm ² AAAC "ALMOND"	30-AL3
Ctrl + o	100mm ² AAAC "OAK"	119-AL3
Ctrl + p	200mm ² AAAC "POPLAR"	239-AL3
Ctrl + r	50mm ² ACSR "RABBIT"	53-AL1/9-ST1A
Ctrl + s	70mm ² ACSR "HORSE"	73-AL1/43-ST1A
Ctrl + t	125mm ² ACSR "TIGER"	131-AL1/31-ST1A
Ctrl + u	300mm ² AAAC "UPAS"	362-AL3
Ctrl + v	(Used elsewhere in the Workbook)	
Ctrl + w	150mm ² ACSR "WOLF"	158-AL1/37-ST1A
Ctrl + x	Copper 16mm²	-
Ctrl + y	Copper 35mm²	-
Ctrl + z	Copper 70mm²	-

Users may develop macros for their own "standard" conductors using the remaining letters of the alphabet. *Note: Ctrl + v has been used elsewhere and may therefore not be used.*

A3 WORKED EXAMPLE

The following worked example considers a design using 100 mm² All Aluminium Alloy Conductor “OAK” at “Normal” altitudes, with a typical downpull not exceeding 1 : 10 and with a basic span of 110 m. In this case, a “2C” weather zone is considered with respect to conductor clash.

A3.1 Go to DESIGN PARAMETERS

The following line parameters and factors of safety have been assumed. Note that the User is able to vary the factors of safety for any component, although this should only be attempted where sufficient knowledge exists of the likely effects. In this instance, a maximum span length of 115 m is assumed.

INPUT	
Line Voltage (kV)	33
Number of Conductors	3
Maximum Wind Span (m)	115
<i>FACTORS of SAFETY:</i>	
Unstayed Pole	2.5
Foundation	2.5
Stayed Pole	2.5
Stays	2.5
Steelwork & Bolts (UTS)	2.5
Pin Insulator	2.5
Equipment Weight (kgf)	0

FIGURE 16

A3.2 Go to SAGNTEN

As indicated in Figure 2, the top spreadsheet in SAGNTEN calculates the basic wind, weight, maximum tension and freezing point tension used by all other spreadsheets for the line design.

The middle spreadsheet in SAGNTEN, as shown in Figure 3, calculates the ordinates for plotting uplift and ground clearance data for a range of conductor temperatures and scaled span lengths. This data may then be used to enable the User to construct his own sag templates for use with hard copy overhead line profiles.

The bottom spreadsheet in SAGNTEN is used to calculate design and / or erection sags and tensions for a User defined range of temperatures and span lengths. The “Design Table” will automatically change to an “Erection Table” if a temperature shift or a percentage over-tension is introduced. In this worked example, Figure 4 indicates a Design Table relating to the selected conductor, and this information has been extracted to Table 2. An Erection Table is copied to Table 3, assuming in this instance, a 10 % overtension to allow for creep compensation.

A3.3 Go to COND SPACING

Using the weather zone maps included in Section 8 of this specification, the User can determine the maximum span length to avoid conductor clash by pressing the appropriate button. The spacing entered for the phase spacing in the SINGLE POLE FIXING and H POLE FIXING spreadsheets must be equal to, or greater than, the spacing entered in the COND SPACING spreadsheet. In Figure 17, it will be seen that the maximum span length must be limited to 120.08 m to ensure that the standard intermediate crossarm spacing of 1200 mm (in respect of crossarm ENATS 439519) is not exceeded in a “2C” weather zone. This is satisfactory for the assumed maximum span limit of 115 m.

Conductor Code Name (if any)	OAK
Weather Index (from ENATS 43-40 maps)	2C
Basic / Recommended Span (m)	110
Freezing Point Tension (kgf)	784.7
Horizontal Conductor Offset or Spacing (mm)	1200
Vertical Conductor Offset or Spacing (mm)	0
Unit Load Reliability Factor	1.00
Spacing Withstand Factor	1.10
Maximum Span to Avoid Clash (m)	120.08
Clash Factor	1.092

FIGURE 17

A3.4 Go to CROSSARM MEMBER

Typical values have been entered for all of the crossarm dimensions, however these are not fixed. Whenever a crossarm member is changed, say from an intermediate/section crossarm to a terminal, the dimensions employed will need to be updated. It is important to check all of the dimensions that relate to the specific crossarm being used *before* considering crossarm hamper suitability.

The facility to alter crossarm member dimensions also permits the User the flexibility to consider sizes (and even steel grade) outside those currently provided within ENATS 43-95.

In the example given below in Figure 18, the standard section size and drillings for intermediate / section and angle crossarms have been included (the section size and drillings for a terminal crossarm are as shown in Figure 12). In the horizontal flange, the centre phase connection hole will only need to be shown when considering single pole

terminal crossarm arrangements. Mild steel has been considered as used for all crossarm arrangements within ENATS 43-95.

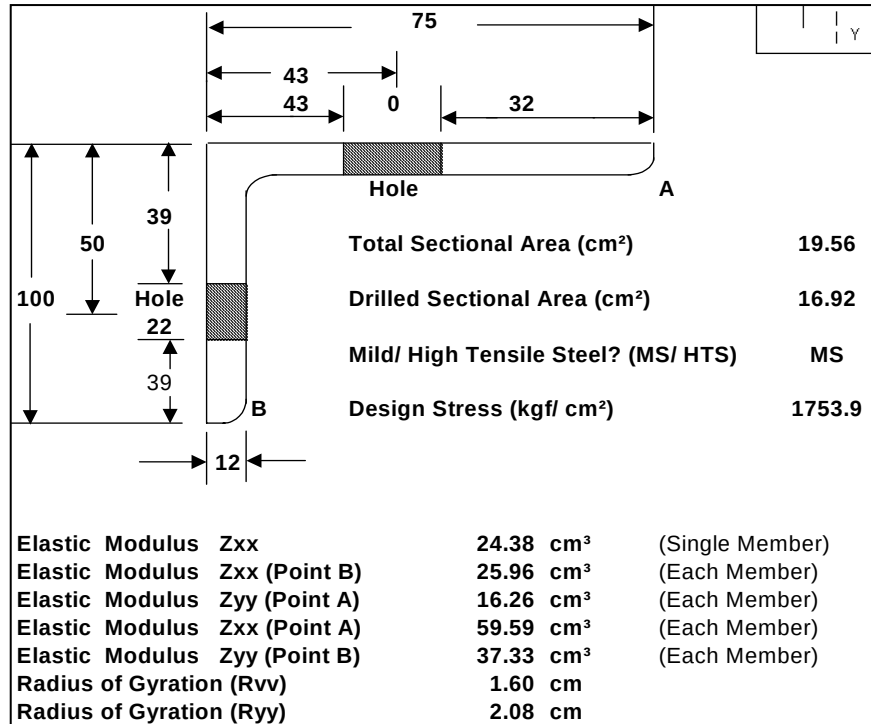


FIGURE 18

A3.5 Go to CROSSARM STRUT

The dimensions shown for the crossarm strut are those specified for use in ENATS 43-95, and these may be altered if the User wishes to consider other strut dimensions.

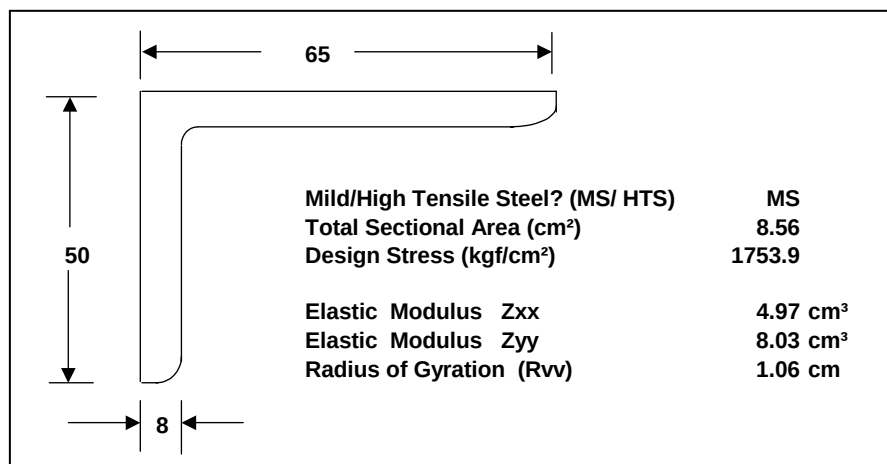


FIGURE 19

A3.6 Go to SINGLE POLE FIXING & H POLE FIXING

These spreadsheets will provide specific design arrangements for each of the crossarm configurations considered. It is important at this stage for the User to consider how and where line span/ line angle data will be stored for future reference. In Figure 20 below, an intermediate crossarm arrangement used as a pin angle, is considered on a single pole.



Having satisfactorily assessed the crossarm arrangement, it is now necessary to review the size of line / terminal strap is required for landing the conductor. This spreadsheet requires the physical dimensions of the strap and the relative position of the strap on the crossarm, based on the worst situation (i.e. smallest pole top diameter). It is important that details of the number and diameter of the fixing holes

are indicated, as this will have a bearing on the strength of the strap. Figure 15 indicates the suitability of the standard 100 mm × 15 mm line strap used with AAAC “OAK” on a Medium wood pole for the worked example.

A3.8 Go to S STAYS & H STAYS

Having established the maximum permissible crossarm parameters from spreadsheets SINGLE POLE FIXING and H POLE FIXING, it is now possible to produce an additional table detailing the maximum permissible angles of deviation/ stay slope each stay type is capable of achieving. This information can be found in S STAYS or H STAYS spreadsheets depending on the Users requirements. For the specific example considered here with “OAK” conductor, Figure 9 indicates the maximum line deviation angles for a range of stay types on a single pole.

Calculations based on the parameters set in the SAGNTEN spreadsheet have allowed the maximum angle of line deviation to be determined relative to each of the stay types available. Should the User require an additional stay loading not provided for in the spreadsheet, a facility has been made available in the “User Defined Stay(s)” cell, where a specific stay strength may be entered. The User must state the specific stay strength required, and the resulting angles of line deviation that the User can achieve will be indicated in the User defined stay angles column.

The information contained in the fields relating to the angles of deviation and stay types can be copied and transferred to an appropriate table for reference. This has been shown in Tables 5 and 6 for both single pole and “H” pole capability relative to “OAK” conductor design.

It can be seen on the spreadsheet that a facility is also available for providing stay spread data relative to the angle of stay considered (see Figure 10). It is important for the User to consider the pole burying depth for each pole used as this figure may change. Reference should be made to the WOODPOLE spreadsheet for typical burying depths. As the information provided by this section of the spreadsheet is not affected by the change in conductor or the loads on the structures, it is recommended that the User produce a table of data for each of the poles considered in his design.

A3.9 Go to STAY FOUND

Having determined maximum angles of line deviation for a number of specific stay types/ strengths, it is now important to assess that the stay foundation is suitable. The STAY FOUND spreadsheet (see Figure 11) requires details of the foundation block being considered and will inform whether the foundation is adequate, depending on stay angle. The spreadsheet also details the minimum stay separation to ensure that the soil frusta are totally separate and do not overlap (i.e. the full soil resistance will be achieved by each stay independently).

A3.10 Go to S STRUT & H STRUT

The next stage is to determine the relative strut loading imposed on the pole structure due to the stresses of the stays. In addition to this, the gradient of the profile is also considered, as the loadings on the pole will increase relative to this value.

Since each component is designed and manufactured in different ways, it is necessary to consider the ultimate strength of the component when applying loads. Due to natural decay and weathering, the original strength of the pole may reduce. To help compensate for this natural degradation and therefore possible reduction in the pole's residual strength, factors of safety are employed.

The factor of safety that is shown on the S STRUT and H STRUT spreadsheets is the "Stayed Pole" value shown in the DESIGN PARAMETERS spreadsheet. The traditional factor of safety employed for wood poles has been 2.5 for normal conditions, and in severe weather environments 3.0 or 3.5. For the specific example using AAAC "OAK" with a factor of safety of 2.5, the pole strut loadings are given in Figure 6.

The maximum gradient normally employed for structure capabilities is 1 : 10. Should the User wish to consider greater gradients than this, he should first check using the SINGLE POLE FIXING or H POLE FIXING spreadsheets. It is important to address the crossarm capabilities before increasing the gradients on the S STRUT or H STRUT spreadsheets.

It should also be noted that when considering the steelwork capabilities severe gradient may be permitted on the steel crossarm, however, this will in turn result in high strut loadings being imposed on the poles and may ultimately be too high for typical pole grades available from the pole supplier.

The information produced in the S STRUT and H STRUT spreadsheet should be compiled to form part of the final specification requirements for the chosen conductor. This information has been copied from the spreadsheet and shown in Tables 7, 8 and 9 for both single and H pole capability relative to AAAC "OAK".

The blue cells contained within the S STRUT and H STRUT spreadsheets once again provide the User with an option to change the stay angles and angles of line deviation for specific situations. It may be necessary for example to determine the strut loading for angles greater than 60°, or optional stay angles other than those provided. Clearly, the stay angles used will mirror those previously considered in the S STAYS and H STAYS spreadsheets.

A3.11 Go to WOODPOLE

The WOODPOLE spreadsheet will provide information for the User relative to the wood poles strength, and will address intermediate pole overturning and foundation capability.

As in other spreadsheets where factors of safety are employed, these are automatically copied from the DESIGN PARAMETERS spreadsheet. As detailed earlier in the text, the User may choose alternative factors of safety as necessary.

Figure 21 shows that the poles are fitted with 2 baulks to aid the foundation stability and the soil type is "ga" Good/ Average. There are a possible nine variations to this spreadsheet, considering different soil conditions (ie. no baulks, one or two baulks in Average / Poor, Good / Average and Good soil conditions). It would assist the User if this data were copied to a separate table as detailed in Table 10 for the AAAC "OAK" worked example.

Design Fibre Stress of Pole and Blocks (N/mm ²)	53.3
Design Average Modulus of Elasticity (N/mm ²)	10054
Wind Pressure (N/m ²)	380
Pole Factor of Safety	2.5
Foundation Factor of Safety	2.5
Number of Conductors	3
Maximum Conductor Pressure (MCP) (kg/m)	1.279
Wind on Pole Considered	y
Average Conductor Level Above Pole Top (mm)	160
Number of Blocks (0, 1[Upper], 2)	2
Top Block Length (mm) (Normal Section 250 x 125mm)	1300 ##Block Length OK##
Groundline to Top Block Pole Bolt (mm)	500
Bottom Block Length (mm) (Normal Section 250 x 125mm)	1300 ##Block Length OK##
Groundline to Bottom Block Pole Bolt (mm)	880
Soil Type (Good, Good/Average, Average/Poor)	ga (Enter G, GA, or AP)

POLE DATA TABLE							
Length (m)	Grade	Top Diam. (mm)	Dia. 1.5m from Butt (mm)	Planting Depth (mm)	Single Pole Strut Strength (kgf)	Maximum Wind Span Length for Specified Pole (m)	Maximum Wind Span Length for Specified Found. (See Key Below)
8.5	MEDIUM	150	215	1600	7768	68	<<<69
9	MEDIUM	150	220	1600	7018	68	<<<68
9.5	MEDIUM	150	225	1600	6395	67	<<<68
10	MEDIUM	150	230	1600	5874	68	<<<68
10.5	MEDIUM	150	235	1600	5431	68	<<<68
11	MEDIUM	150	240	1600	5051	68	<<<69
11.5	MEDIUM	150	245	1800	4853	68	<<<69
12	MEDIUM	150	250	1800	4550	69	<<<69
13	MEDIUM	160	260	2100	4814	70	<<<71
14	MEDIUM	160	275	2100	4476	75	<<<76
15	MEDIUM	165	290	2100	4488	81	<<<81
16	MEDIUM	170	305	2400	4645	87	<<<87
17	MEDIUM	180	320	2400	4979	94	<<<95
18	MEDIUM	180	330	2400	4623	95	<<<95
20	MEDIUM	180	360	3000	4468	103	<<<104
22	MEDIUM	190	380	3000	4455	109	<<<109
8.5	STOUT	190	265	1600	19025	134	<<<135
9	STOUT	190	275	1600	17629	140	>>>>>> 134
9.5	STOUT	190	280	1600	15932	138	>>>>>> 126
10	STOUT	190	285	1600	14513	136	>>>>>> 119
10.5	STOUT	190	290	1600	13313	135	>>>>>> 112
11	STOUT	190	295	1600	12286	134	>>>>>> 106
11.5	STOUT	190	300	1800	11725	133	<<<134
12	STOUT	190	305	1800	10915	133	<<<133
13	STOUT	195	320	2100	10815	139	<<<139
14	STOUT	195	335	2100	9866	144	<<<144
15	STOUT	195	350	2100	9113	148	<<<148
16	STOUT	200	365	2400	9192	154	<<<154
17	STOUT	200	375	2400	8402	153	<<<153
18	STOUT	200	390	2400	7946	156	<<<157
20	STOUT	200	415	3000	7323	158	<<<158
22	STOUT	200	435	3000	6468	155	<<<156
24	STOUT	200	470	3000	6238	164	<<<164

FIGURE 21

The WOODPOLE spreadsheet is extremely important to the User and ultimately to the design specification being produced. Two main points are detailed as follows:

- (i) The figures given in the "Single Pole Strut Strength" column are essential when pole choice is required. Under no circumstances should a pole be selected for use in the design if the strut loadings indicated in the S STRUT or H STRUT spreadsheets exceed the values shown in the WOODPOLE spreadsheet. Where "H" pole supports are considered, the pole top diameter is recommended to be a minimum of 175 mm for

Medium-grade poles or 210 mm Stout-grade poles. In this example, the common minimum pole top diameters have been used.

- (ii) It should be noted that the two columns headed “Maximum Wind Span Length for Specified Pole” and “Maximum Wind Span Length for Specified Found” detail independent values relating to the intermediate / section pole. For a specified pole length and buried length, the span length will not change and should never be exceeded. The foundation capability will change whenever the soil type or baulk dimensions vary. The span lengths derived for the specified foundation may be less than that shown in the “Maximum Wind Span Length for Specified Pole” column, the lower of the two values must always be adopted for the design specification and arrows on the WOODPOLE spreadsheet indicate this.

A3.11 Go to STRUT FOUND

This spreadsheet provides information relating to the pole’s resistance to sinking when a strut load is imposed. Both single and H poles are assessed individually, with the User detailing specific foundation block information. Unlike the other spreadsheets this exercise need only be carried out once and tabulated for future reference, as this data is not conductor specific. However, should other sizes of poles be considered in the WOODPOLE spreadsheet, then the User must change the relative foundation blocking fields contained in this spreadsheet to update his design data. Note that the pole dimension data is copied from the WOODPOLE spreadsheet and therefore does not need amendment.

A3.12 DETAILING of RESULTS

The information necessary to carry out an overhead line design has now been compiled by approaching each of the spreadsheets in turn from the Workbook. The following Tables produced show this information in typical form and can be used in a design specification issued to overhead line staff. This information can also be provided and entered into computerised design packages for profiling and scheduling.

Tension Reduction at 15°C (%)	0
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Table 1: Ordinates for Plotting Ground Clearance Templates

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	865.7	0.17	0.23	0.30	0.38	0.47	0.57	0.68	0.80	0.93
0	784.7	0.19	0.26	0.33	0.42	0.52	0.63	0.75	0.88	1.02
5	715.2	0.21	0.28	0.37	0.46	0.57	0.69	0.82	0.97	1.12
10	649.1	0.23	0.31	0.40	0.51	0.63	0.76	0.91	1.07	1.24
15	587.2	0.25	0.34	0.45	0.56	0.70	0.84	1.00	1.18	1.37
20	530.3	0.28	0.38	0.49	0.62	0.77	0.93	1.11	1.30	1.51
25	479.0	0.31	0.42	0.55	0.69	0.85	1.03	1.23	1.44	1.67
30	433.7	0.34	0.46	0.60	0.76	0.94	1.14	1.36	1.59	1.85
35	394.4	0.37	0.51	0.66	0.84	1.04	1.26	1.49	1.75	2.03
40	360.7	0.41	0.56	0.73	0.92	1.13	1.37	1.63	1.92	2.22
45	332.0	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	2.42
50	307.6	0.48	0.65	0.85	1.08	1.33	1.61	1.92	2.25	2.61
55	286.8	0.51	0.70	0.91	1.16	1.43	1.73	2.05	2.41	2.80
60	268.9	0.55	0.75	0.97	1.23	1.52	1.84	2.19	2.57	2.98
65	253.5	0.58	0.79	1.03	1.31	1.61	1.95	2.32	2.73	3.16
70	240.2	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.88	3.34
75	228.4	0.64	0.88	1.15	1.45	1.79	2.17	2.58	3.03	3.51
80	218.1	0.68	0.92	1.20	1.52	1.88	2.27	2.70	3.17	3.68

Table 2: Sag / Tension Design Table

Conductor Code Name	OAK
Basic / Recommended Span (m)	110
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	10.0 at 15°C
Required Percentage Increase in Tension (%)	10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	936.3	0.16	0.21	0.28	0.35	0.44	0.53	0.63	0.74	0.86
0	853.2	0.17	0.23	0.31	0.39	0.48	0.58	0.69	0.81	0.94
5	781.2	0.19	0.26	0.34	0.42	0.52	0.63	0.75	0.89	1.03
10	711.9	0.21	0.28	0.37	0.47	0.57	0.70	0.83	0.97	1.13
15	645.9	0.23	0.31	0.41	0.51	0.63	0.77	0.91	1.07	1.24
20	584.3	0.25	0.34	0.45	0.57	0.70	0.85	1.01	1.18	1.37
25	527.6	0.28	0.38	0.50	0.63	0.78	0.94	1.12	1.31	1.52
30	476.6	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.45	1.68
35	431.6	0.34	0.46	0.61	0.77	0.95	1.15	1.36	1.60	1.86
40	392.6	0.38	0.51	0.67	0.84	1.04	1.26	1.50	1.76	2.04
45	359.2	0.41	0.56	0.73	0.92	1.14	1.38	1.64	1.92	2.23
50	330.7	0.45	0.61	0.79	1.00	1.24	1.50	1.78	2.09	2.42
55	306.5	0.48	0.65	0.85	1.08	1.33	1.62	1.92	2.26	2.62
60	285.8	0.52	0.70	0.92	1.16	1.43	1.73	2.06	2.42	2.81
65	268.1	0.55	0.75	0.98	1.24	1.53	1.85	2.20	2.58	2.99
70	252.8	0.58	0.79	1.04	1.31	1.62	1.96	2.33	2.73	3.17
75	239.5	0.61	0.84	1.09	1.38	1.71	2.07	2.46	2.89	3.35
80	227.9	0.65	0.88	1.15	1.45	1.80	2.17	2.59	3.03	3.52

Table 3: Sag / Tension Erection Table

Structure type	100mm ² AAAC "OAK"	Structure drawing no.
Intermediate through & pin angle, single pole, single crossarm (10 kN post / pin insulators)	up to 8° (33 kV) up to 11° (11 kV)	434003
Intermediate through & pin angle, single pole, double crossarm (10 kN post / pin insulators)	up to 11° (33 kV) up to 11° (11 kV)	434004
Angle / Section, Single pole, double crossarm	up to 60°	434005
Angle / Section, "H" pole, double crossarm	up to 60°	434006
Angle / Section, "H" pole, double crossarm (Occasional long spans)	0°; 155 m 60°; 144 m (Limited by clash)	434007
Tee-Off, Single pole, double crossarm (Un-braced)	Not suitable	434008
Terminal, Single pole, double crossarm (Un-braced)	Not suitable	434009
Terminal, Single pole, double crossarm (Braced)	OK	434010
Terminal, "H" pole, double crossarm	OK	434011
Intermediate through and pin angle, single pole, single crossarm (Renovation crossarms – 10 kN pins)	up to 5° (33 kV) 109 m up to 11° (11 kV) 109 m (Limited by clash)	434012

Table 4: Crossarm / Structure Suitability

Notes: The Table above assumes "OAK" basic span 110 m, with clash span 115 m (except where otherwise shown), "Normal" altitudes, and maximum gradient for downpull 1 : 10. Mild steel crossarm and bolts are assumed.

*Insulator heights are taken as 450 mm for 33 kV and 260 mm for 11 kV
Insulator weights are taken as 10 kgf for pin / posts; 37 kgf for tension strings.
Maximum line deviation angles are shown where applicable.*

Angle of Stay Slope	MAXIMUM ANGLE of LINE DEVIATION								User Defined Stay(s) No. 1 6281
	Grade 700 Staywire				Grade 1150 Staywire				
	1x7/4.00	2 x 7/4.00	1x19/3.55	2x19/3.55	1x7/3.25	2 x 7/3.25	1x7/4.00	2x7/4.00	
	Stay Strength (kgf)				Stay Strength (kgf)				
	6281	6281	11216	11216	6628	6628	10299	10299	
30	10	27	24	57	11	29	22	51	10
35	13	33	29	60	14	35	26	60	13
40	15	38	34	60	16	41	30	60	15
45	17	44	38	60	19	47	34	60	17

Table 5: Single Pole Stay Capability – 100 mm² AAAC “OAK”

Angle of Stay Slope	MAXIMUM ANGLE of LINE DEVIATION								User Defined Stay(s) No. 1 6281
	Grade 700 Staywire				Grade 1150 Staywire				
	3x7/4.00	4 x 7/4.00	3x19/3.55	4x19/3.55	3x7/3.25	4 x 7/3.25	3x7/4.00	4x7/4.00	
	Stay Strength (kgf)				Stay Strength (kgf)				
	6281	6281	11216	11216	6628	6628	10299	10299	
30	46	60	60	60	49	60	60	60	9
35	55	60	60	60	59	60	60	60	12
40	60	60	60	60	60	60	60	60	14
45	60	60	60	60	60	60	60	60	17

Table 6: “H” Pole Stay Capability – 100 mm² AAAC “OAK”

STRUT LOAD in POLE with ONE or TWO STAYS (kgf)				
Line Angle (Degrees)	Stay Angle (Degrees)			
	30	35	40	45
Level Conditions				
0	3962	3559	3245	2990
5	5473	4804	4284	3863
10	6977	6045	5319	4731
15	8472	7277	6348	5594
20	9955	8500	7368	6450
25	11422	9710	8378	7297
30	12872	10905	9375	8134
35	14301	12084	10359	8960
40	15707	13243	11326	9771
45	17087	14381	12275	10568
50	18438	15495	13205	11348
55	19758	16583	14113	12110
60	21044	17644	14999	12853
Downpull 1: 10				
0	5954	5551	5237	4982
5	7465	6796	6276	5855
10	8969	8037	7311	6723
15	10464	9269	8340	7586
20	11947	10492	9360	8442
25	13414	11702	10370	9290
30	14864	12897	11367	10127
35	16293	14076	12351	10952
40	17699	15235	13318	11763
45	19079	16373	14268	12560
50	20430	17487	15197	13340
55	21750	18575	16106	14102
60	23037	19636	16991	14845

Table 7: Pole Strut Load (kgf) – 1 or 2 stays – 100 mm² “OAK”

STRUT LOAD in POLE with THREE STAYS (kgf)				
(60% of Line Load for Inner Pole - Worst Case)				
Line Angle (Degrees)	Stay Angle (Degrees)			
	30	35	40	45
Level Conditions				
0	2328	2065	1861	1695
5	3234	2813	2485	2219
10	4137	3557	3106	2740
15	5034	4296	3723	3258
20	5923	5030	4335	3771
25	6804	5756	4941	4280
30	7674	6473	5539	4782
35	8531	7180	6129	5277
40	9375	7876	6710	5764
45	10202	8558	7279	6242
50	11013	9227	7837	6710
55	11805	9880	8382	7167
60	12577	10516	8913	7613
Downpull 1: 10				
0	3324	3061	2857	2691
5	4230	3809	3481	3215
10	5133	4553	4102	3736
15	6030	5292	4719	4254
20	6919	6026	5331	4767
25	7800	6752	5937	5276
30	8670	7469	6535	5778
35	9527	8176	7125	6273
40	10371	8872	7706	6760
45	11199	9554	8275	7238
50	12009	10223	8833	7706
55	12801	10876	9378	8163
60	13573	11512	9909	8609

**Table 8: Pole Strut Load (kgf) – 3 stays – 100 mm² “OAK”
(60% of line load for inner pole – worst case)**

STRUT LOAD in POLE with FOUR STAYS (kgf) (50% of Line Load for Each Pole)				
Line Angle (Degrees)	Stay Angle (Degrees)			
	30	35	40	45
Level Conditions				
0	2078	1859	1689	1551
5	2834	2482	2209	1987
10	3586	3102	2727	2422
15	4333	3719	3241	2853
20	5075	4330	3751	3281
25	5808	4935	4256	3705
30	6533	5533	4755	4123
35	7248	6122	5246	4536
40	7951	6702	5730	4942
45	8641	7270	6205	5340
50	9316	7828	6669	5730
55	9976	8372	7124	6111
60	10619	8902	7566	6483
Downpull 1: 10				
0	3074	2856	2685	2547
5	3830	3478	3205	2983
10	4582	4099	3723	3418
15	5329	4715	4237	3849
20	6071	5326	4747	4277
25	6804	5931	5252	4701
30	7529	6529	5751	5119
35	8244	7118	6242	5532
40	8947	7698	6726	5938
45	9637	8266	7201	6336
50	10312	8824	7666	6726
55	10972	9368	8120	7107
60	11615	9898	8562	7479

**Table 9: Pole Strut Load (kgf) – 4 stays – 100 mm² “OAK”
(50 % of line load for each pole)**

Pole Data					Single Pole Strut Strength (kgf)	Maximum Span Length (m) for Specified Pole and Foundation								
Length (m)	Grade	Top	Dia.	Planting Depth		Average-Poor Soil			Good-Average Soil			Good Soil		
		Diam. (mm)	1.5m from Butt				No Block	Single Block	Double Block	No Block	Single Block	Double Block	No Block	Single Block
8.5	MEDIUM	150	215	1600	7768	24	55	68	39	68	68	55	68	68
9	MEDIUM	150	220	1600	7018	22	51	68	37	68	68	52	68	68
9.5	MEDIUM	150	225	1600	6395	20	48	67	35	67	67	49	67	67
10	MEDIUM	150	230	1600	5874	19	45	68	33	68	68	47	68	68
10.5	MEDIUM	150	235	1600	5431	17	42	68	31	68	68	44	68	68
11	MEDIUM	150	240	1600	5051	16	39	65	29	64	68	42	68	68
11.5	MEDIUM	150	245	1800	4853	26	54	68	44	68	68	63	68	68
12	MEDIUM	150	250	1800	4550	24	51	69	42	69	69	60	69	69
13	MEDIUM	160	260	2100	4814	42	70	70	69	70	70	70	70	70
14	MEDIUM	160	275	2100	4476	39	67	75	66	75	75	75	75	75
15	MEDIUM	165	290	2100	4488	36	62	81	62	81	81	81	81	81
16	MEDIUM	170	305	2400	4645	59	87	87	87	87	87	87	87	87
17	MEDIUM	180	320	2400	4979	56	83	94	94	94	94	94	94	94
18	MEDIUM	180	330	2400	4623	52	78	95	89	95	95	95	95	95
20	MEDIUM	180	360	3000	4468	103	103	103	103	103	103	103	103	103
22	MEDIUM	190	380	3000	4455	105	109	109	109	109	109	109	109	109
8.5	STOUT	190	265	1600	19025	29	59	92	48	93	134	67	127	134
9	STOUT	190	275	1600	17629	27	55	86	46	88	134	65	121	140
9.5	STOUT	190	280	1600	15932	25	51	80	43	83	126	61	114	138
10	STOUT	190	285	1600	14513	23	48	75	40	78	119	58	107	136
10.5	STOUT	190	290	1600	13313	21	45	71	38	73	112	55	102	135
11	STOUT	190	295	1600	12286	19	42	66	36	69	106	52	96	134
11.5	STOUT	190	300	1800	11725	31	58	94	54	93	133	76	129	133
12	STOUT	190	305	1800	10915	29	55	89	51	89	133	73	123	133
13	STOUT	195	320	2100	10815	51	81	128	85	129	139	118	139	139
14	STOUT	195	335	2100	9866	47	74	118	80	120	144	113	144	144
15	STOUT	195	350	2100	9113	44	68	108	75	112	148	107	148	148
16	STOUT	200	365	2400	9192	71	99	148	117	154	154	154	154	154
17	STOUT	200	375	2400	8402	66	92	138	110	149	153	153	153	153
18	STOUT	200	390	2400	7946	62	86	129	106	142	156	149	156	156
20	STOUT	200	415	3000	7323	134	158	158	158	158	158	158	158	158
22	STOUT	200	435	3000	6468	122	148	155	155	155	155	155	155	155
24	STOUT	200	470	3000	6238	115	137	164	164	164	164	164	164	164

**Table 10: Wind Loading Spans, Pole Strut Strengths and Foundations
100 mm² AAC “OAK”**